

SHETH MVLU COLLEGE

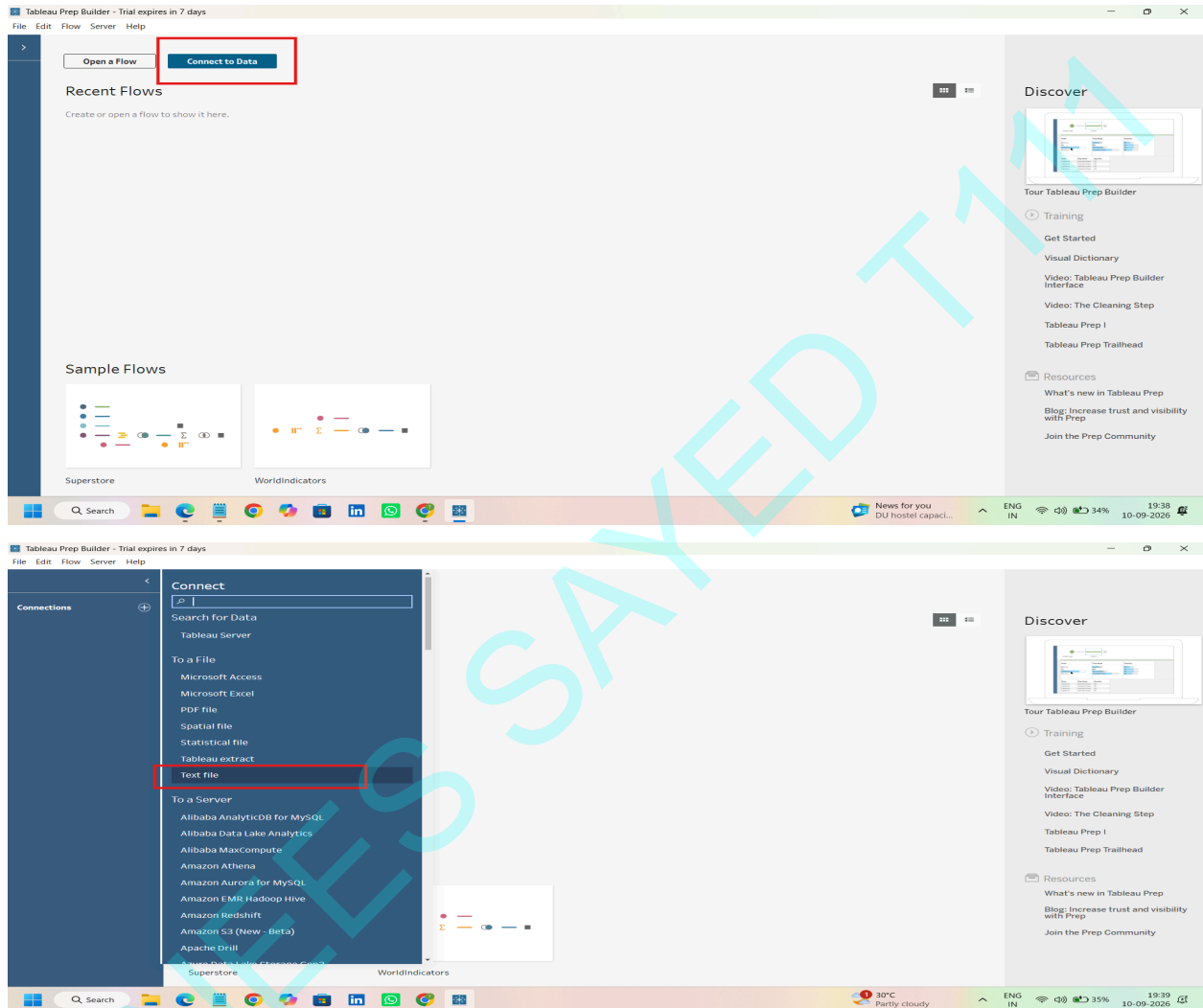
SUBJECT:- TABLEAU

Practical 10

Aim:- Data Preparation and Transformation using Tableau Prep

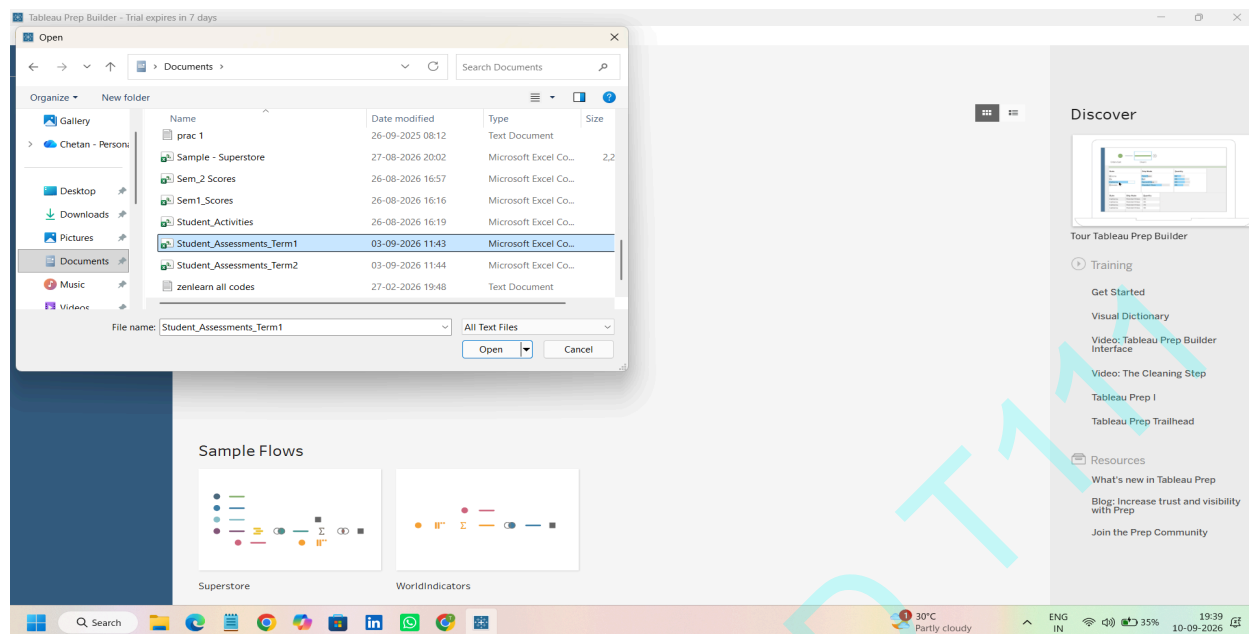
A: Connect and Load Data into Tableau Prep

Step 1:- Launch Tableau Prep Builder And Click The Connect To Data Icon And Select The CSV File Student_Assessments_Term1.csv

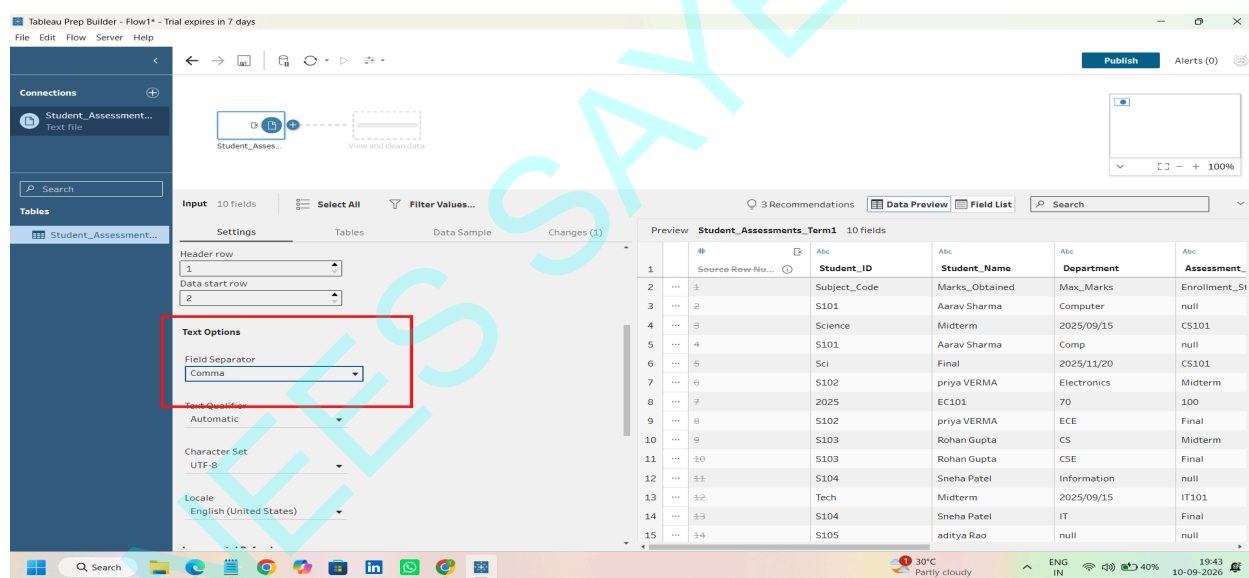


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Step 2:- In the bottom Input configuration pane, verify: o Field Separator: Comma (,) First row contains header: Checked



Step 3:- In the top-left Connections pane, click the + (Add connection) button again. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open. Click + (Add connection) once more select Text file choose Course_Credit_Master.csv click Open.

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Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Connections

Student_Assessment...
Text File

Tables

Student_Assessment...

Input 10 fields

Select All Filter Values...

Settings

Header row 1

Data start row 2

Text Options

Field Separator Comma

Text Qualifier Automatic

Character Set UTF-8

Locale English (United States)

Preview Student_Assessments_Term1 10 fields

1	Source Row Number	Student_ID	Student_Name	Department	Assessment
2	1	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_St
3	2	S101	Aarav Sharma	Computer	null
4	3	Science	Midterm	2025/09/15	CS101
5	4	S101	Aarav Sharma	Comp	null
6	5	Sci	Final	2025/11/20	CS101
7	6	S102	priya VERMA	Electronics	Midterm
8	7	2025	EC101	70	100
9	8	S102	priya VERMA	ECE	Final
10	9	S103	Rohan Gupta	CS	Midterm
11	10	S103	Rohan Gupta	CSE	Final
12	11	S104	Sneha Patel	Information	null
13	12	Tech	Midterm	2025/09/15	IT101
14	13	S104	Sneha Patel	IT	Final
15	14	S105	aditya Rao	null	null

Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Open

Documents

Search Documents

Organize New folder

Chetan - Person

Desktop

Downloads

Pictures

Documents

Music

Videos

Files

Name

Date modified

Type

Size

Sample - Superstore

Sem_2_Scores

Sem1_Scores

Student_Activities

Student_Assessments_Term1

Student_Assessments_Term2

zenlearn all codes

zomato

File name: Student_Assessments_Term2

All Text Files

Open Cancel

Field Separator Comma

Text Qualifier Automatic

Character Set UTF-8

Locale English (United States)

Preview Student_Assessments_Term1 10 fields

1	Source Row Number	Student_ID	Student_Name	Department	Assessment
2	1	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_St
3	2	S101	Aarav Sharma	Computer	null
4	3	Science	Midterm	2025/09/15	CS101
5	4	S101	Aarav Sharma	Comp	null
6	5	Sci	Final	2025/11/20	CS101
7	6	S102	priya VERMA	Electronics	Midterm
8	7	2025	EC101	70	100
9	8	S102	priya VERMA	ECE	Final
10	9	S103	Rohan Gupta	CS	Midterm
11	10	S103	Rohan Gupta	CSE	Final
12	11	S104	Sneha Patel	Information	null
13	12	Tech	Midterm	2025/09/15	IT101
14	13	S104	Sneha Patel	IT	Final
15	14	S105	aditya Rao	null	null

Tableau Prep Builder - Flow1* - Trial expires in 7 days

File Edit Flow Server Help

Connections

Student_Assessment...
Text File

Student_Assessment...
Text File

Course_Credit_Mast...
Text File

Tables

Course_Credit_Master

Input 5 fields

Select All Filter Values...

Settings

Header row 1

Data start row 2

Text Options

Field Separator Comma

Text Qualifier Automatic

Character Set UTF-8

Preview Course_Credit_Master 5 fields

1	Source Row Number	Subject_Code	Subject_Title	Credit_Hours	Passing_Marks
2	1	CS101	Data Structures	4	40
3	2	CS201	Algorithms	4	40
4	3	EC101	Digital Logic	3	40
5	4	EC201	Signals & Systems	4	40
6	5	IT101	Database Systems	4	40
7	6	IT201	Cloud Computing	4	40
8	7	ME101	Thermodynamics	3	40

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Step 4:- On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon. 2. Click the small + icon that appears to its right. Select Clean Step (the step with the cleaning tool icon). The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.

The first screenshot shows the Tableau Prep Builder interface with the 'Clean' step selected in the flow canvas. The 'Profile Pane' is open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1. The second screenshot shows the 'Clean 1' step selected in the flow canvas, with the 'Profile Pane' displaying the same data.

Student_ID	Student_Name	Department	Assessment...	Exam_Date	F6	F7
2025	Aarav Sharma	Computer	2025/09/15	20-11-2025	100	100
S101	aditya Rao	Computer	2025/09/15	2025.09.15	35	45
S102	Ananya Singh	Computer	2025/11/20	2025.09.15	CS101	52
S103	EC101	Computer	Final	2025.11.20	EC101	65
S104	Kavya Beddy	Computer	Enrollment_Status	2025/09/15	IT101	78
S105	Marks_Obtained	CS	Final	2025/11/20	40	92
S106	Mechanical	CSE	Final	40	72	94
S107	Midterm	ECE	ME101	72	85	96
S108	priya VERMA	Electronics	Info	85	88	Active
S109	Rahul Nair	Info	Midterm	90	90	Droppe
Sci	Rohan Gupta	Information				

B: Clean and Profile Data

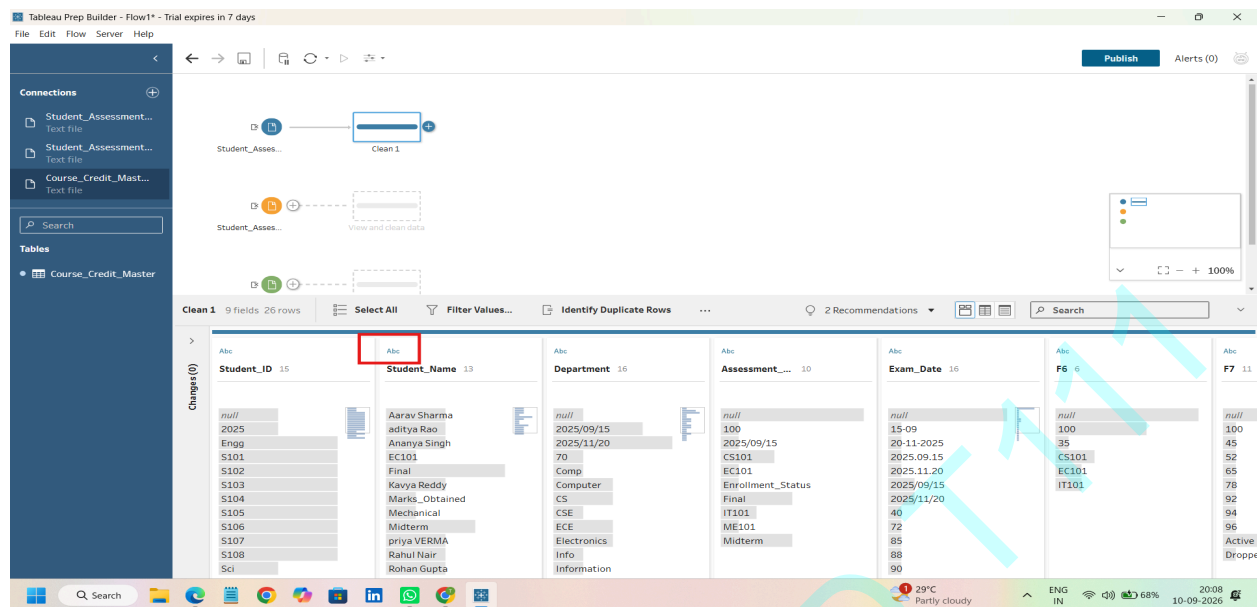
Step 1:- Click on the Clean 1 step connected to Student_Assessments_Term1.

2. Look at the cards displayed in the Profile Pane:

- o Each card represents one column.
- o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., 'Aarav Sharma' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao').
- o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/, -, .).

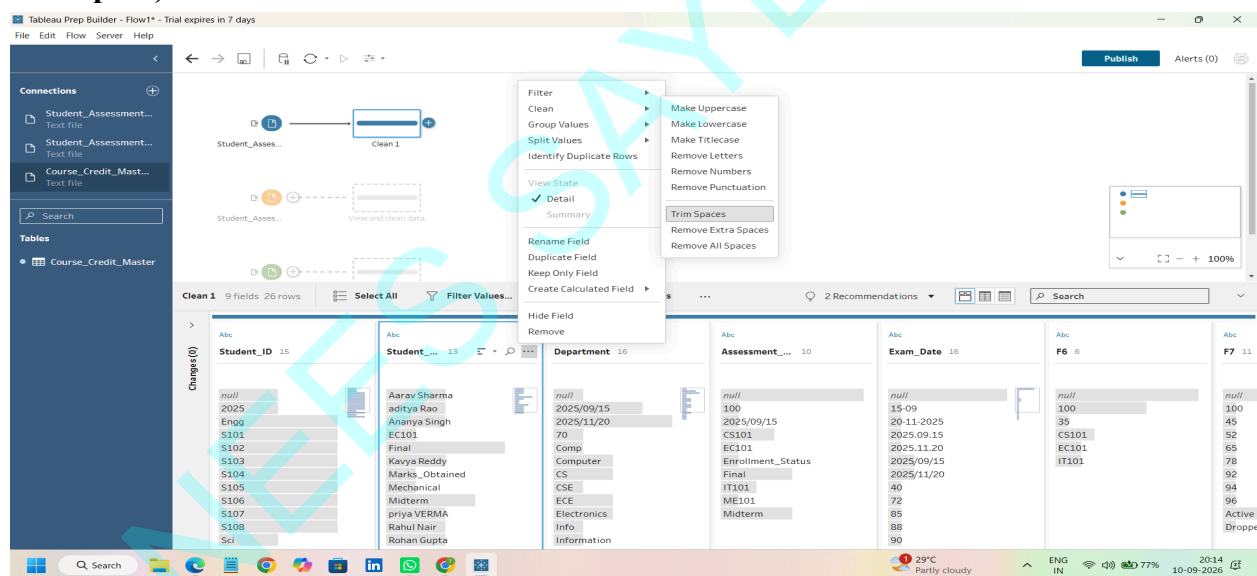
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Step 2:- On the Student_Name profile card:

- o Click the More options (...) icon at the top right of the card.
- o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).



Click More options (...) again --> Clean --> select Make Uppercase (or Make Lowercase / standard proper formatting).

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The screenshots show the Tableau Prep Builder interface. The top screenshot displays a context menu for the 'Exam_Date' column, with options such as 'Make Uppercase', 'Make Lowercase', 'Make Titlecase', 'Remove Letters', 'Remove Numbers', 'Remove Punctuation', 'Trim Spaces', 'Remove Extra Spaces', and 'Remove All Spaces'. The bottom screenshot shows the 'Exam_Date' column with a red box highlighting the data type icon (a calendar icon) above the column name, indicating the step to change the data type to Date.

Step 3:- Locate the Exam_Date profile card. Click the data type icon (currently showing Abc) directly above the column name. Select Date from the dropdown menu. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09 2025, 2025.11.20) into standardized calendar dates.

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The image displays two screenshots of the Tableau Prep Builder interface. The top screenshot shows a workflow with 'Clean 1' applied to 'Student_Asses...'. The bottom screenshot shows 'Clean 2' applied to the same data, with a new column 'Exam Date' added and highlighted by a red box. The 'Exam Date' column contains values such as '01/01/1972', '01/01/1985', '01/01/1988', '01/01/1990', '01/01/1992', '01/01/1998', '15/09/2025', '20/11/2025', and '01/01/2040'.

- Step 4:-** On the Flow canvas, hover over the Student_Assessments_Term2 input icon.
- Click + --> select Clean Step (creates Clean 2).
 - On the Full_Name card in Clean 2:
 - Click More options (...) --> Clean --> Trim Spaces.

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Tableau Prep Builder - Flow1* - Trial expires in 7 days

Connections:

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables:

- Course_Credit_Master

Input: 4 fields

Settings:

- + Clean Step
- New Rows
- Aggregate
- Pivot
- Join
- Union
- Script
- Prediction
- Output
- Insert Flow

Connection Details:

Text file

File: C:\Users\CHETAN\Documents\Student_Assessments_Term2.csv

Header Options:

- Generate header (ex: F1, F2, F3)
- Set header

Header and Data Start Row:

Preview: Student_Assessments_Term2 4 fields

Source Row Number	F1	F2	F3
1	Std_ID,Full_Name,De...	null	null
2	core>Total_Possible,E...	null	null
3	S101,Aarav	Sharma,Computer	null
4	Science,Midterm,2026...	null	null
5	S101,Aarav	Sharma,CS,Final,2026...	null
6	S102,Priya	Verma,ECE,Midterm,2...	null
7	S102,Priya	null	null
8	Verma,Electronics,Fin...	null	null
9	S103,Rohan	Gupta,CSE,Midterm,2...	null
10			

Tableau Prep Builder - Flow1* - Trial expires in 7 days

Connections:

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables:

- Course_Credit_Master

Input: 4 fields

Settings:

- Filter
- Clean
- Group Values
- Split Values
- Identify Duplicate Rows
- View State
- Detail
- Summary
- Rename Field
- Duplicate Field
- Keep Only Field
- Create Calculated Field
- Hide Field
- Remove

Connection Details:

Text file

File: C:\Users\CHETAN\Documents\Student_Assessments_Term2.csv

Header Options:

- Generate header (ex: F1, F2, F3)
- Set header

Header and Data Start Row:

Preview: Identify Duplicate Rows

Source Row Number	F1	F2	F3
1	core>Total_Possible,Enrol	null	null
2	Nair,Electronics,Final,202	Gupta,CS,Final,2026/04/2	Gupta,CSE,Midterm,2026
3	S101,Aarav	Nair,ECE,Midterm,2026/0	Patel,Information
4	S102,Priya	Patel,IT,Midterm,2026/02	Reddy,IT,Final,2026/04/2
5	S103,Rohan	Reddy,IT,Midterm,2026/0	Sharma,Computer
6	S104,Sneha	Sharma,CS,Final,2026/04	Singh,Computer
7	S106,Ananya	Singh,CSE,Midterm,2026/0	
8	S107,Rahul		
9	S108,Kavya		
10	Science,Final,2026/04/25		
11	Science,Midterm,2026/02		
12	Std_ID,Full_Name,Dept,E		

Tableau Prep Builder - Flow1* - Trial expires in 7 days

Connections:

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Tables:

- Course_Credit_Master

Input: 4 fields

Settings:

- Filter
- Clean
- Group Values
- Split Values
- Identify Duplicate Rows
- View State
- Detail
- Summary
- Rename Field
- Duplicate Field
- Keep Only Field
- Create Calculated Field
- Hide Field
- Remove

Connection Details:

Text file

File: C:\Users\CHETAN\Documents\Student_Assessments_Term2.csv

Header Options:

- Generate header (ex: F1, F2, F3)
- Set header

Header and Data Start Row:

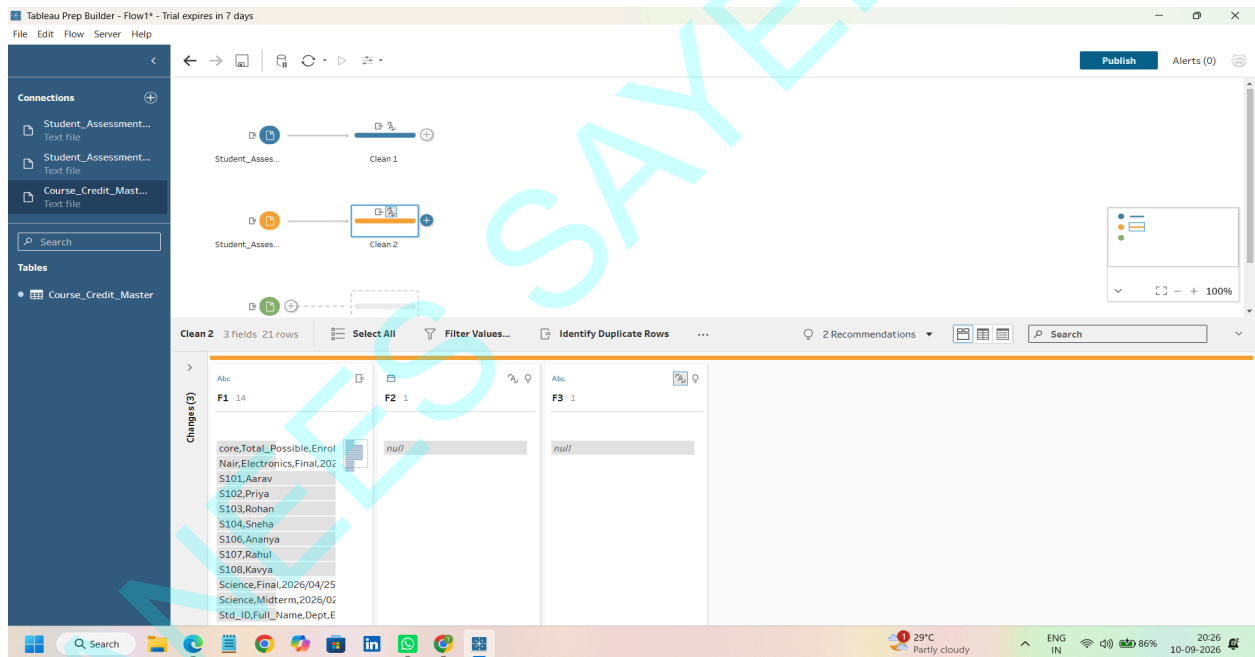
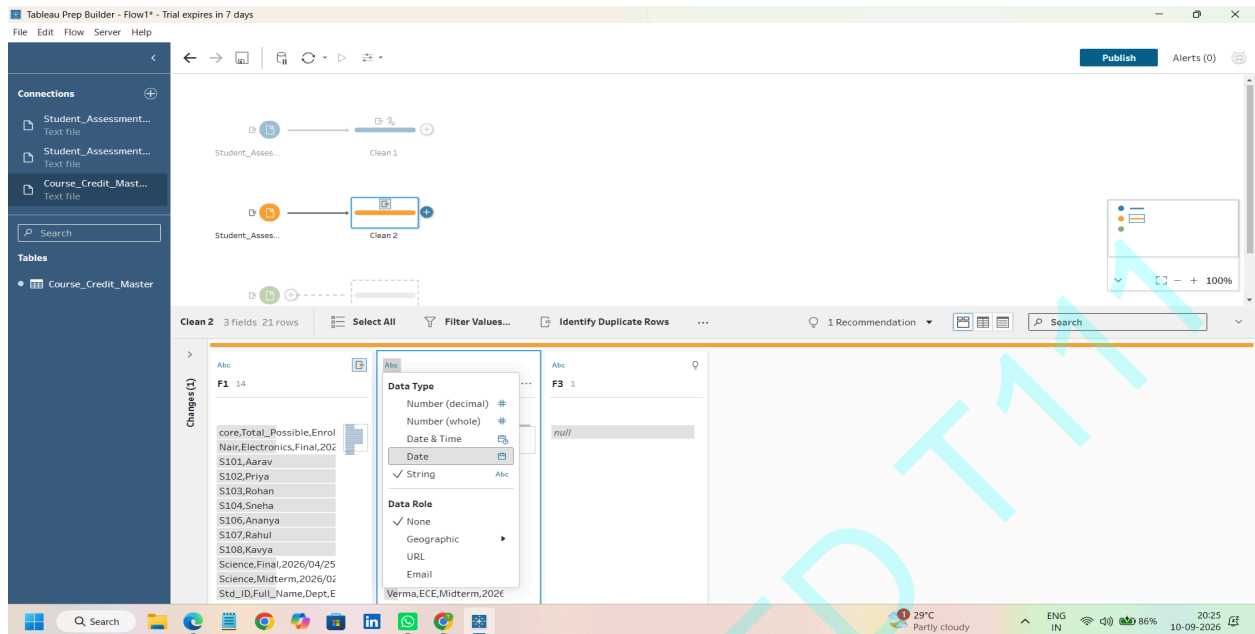
Preview: Clean 2

Source Row Number	F1	F2	F3
1	core>Total_Possible,Enrol	null	null
2	Nair,Electronics,Final,202	Gupta,CS,Final,2026/04/2	Gupta,CSE,Midterm,2026
3	S101,Aarav	Nair,ECE,Midterm,2026/0	Patel,Information
4	S102,Priya	Patel,IT,Midterm,2026/02	Reddy,IT,Final,2026/04/2
5	S103,Rohan	Reddy,IT,Midterm,2026/0	Sharma,Computer
6	S104,Sneha	Sharma,CS,Final,2026/04	Singh,Computer
7	S106,Ananya	Singh,CSE,Midterm,2026/0	
8	S107,Rahul		
9	S108,Kavya		
10	Science,Final,2026/04/25		
11	Science,Midterm,2026/02		
12	Std_ID,Full_Name,Dept,E		

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On the Assessment_Date card: o Click the Abc icon --> change the data type to Date.



C: Group and Standardize Values

Step 1:- Click on the Clean 1 step connected to Student_Assessments_Term1.

2. Locate the Department card in the Profile Pane.

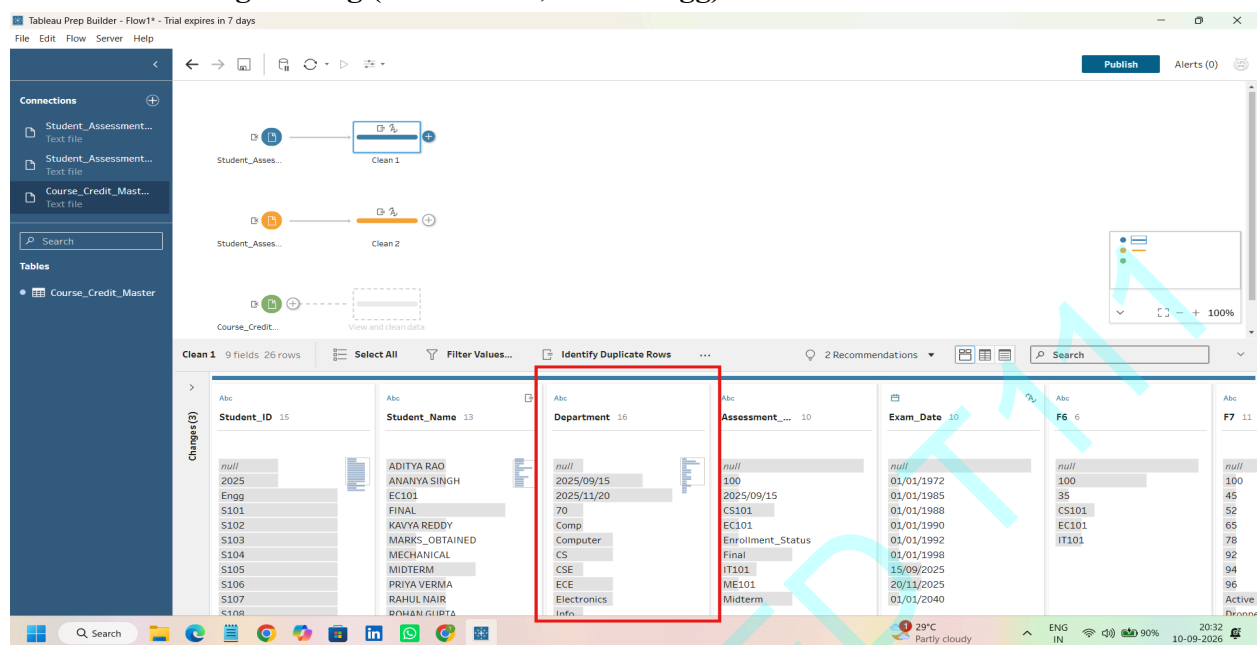
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:

- o Computer Science (Computer Science, Comp Sci, CS, CSE)
- o Electronics (Electronics, ECE, Electronics Engg)
- o Information Technology (Information Tech, IT, Info Tech)

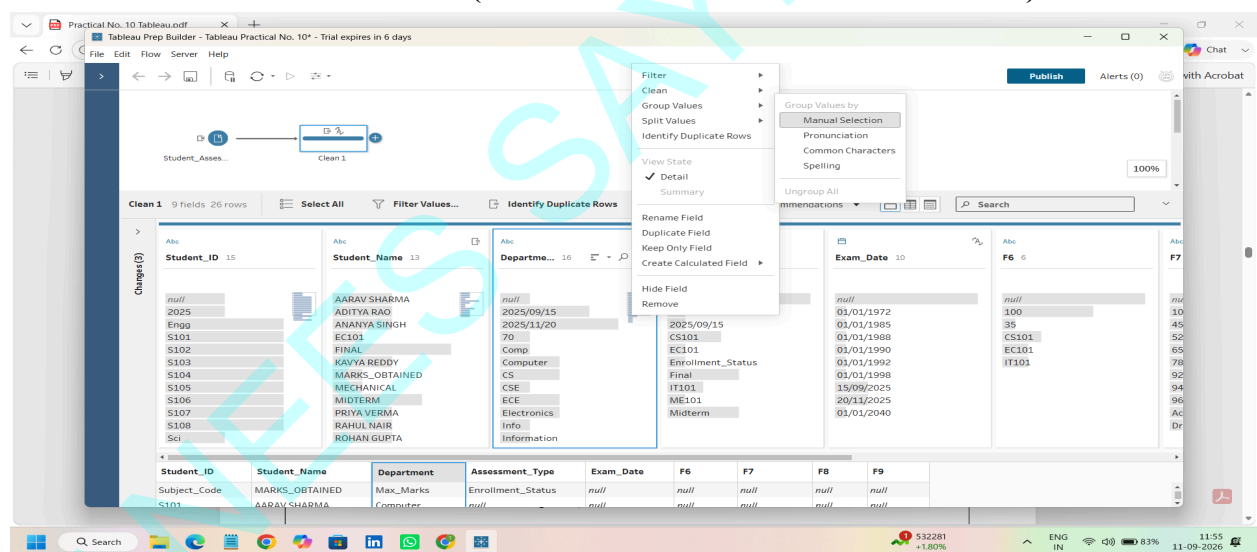
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o Mechanical Engineering (Mechanical, Mech Engg)

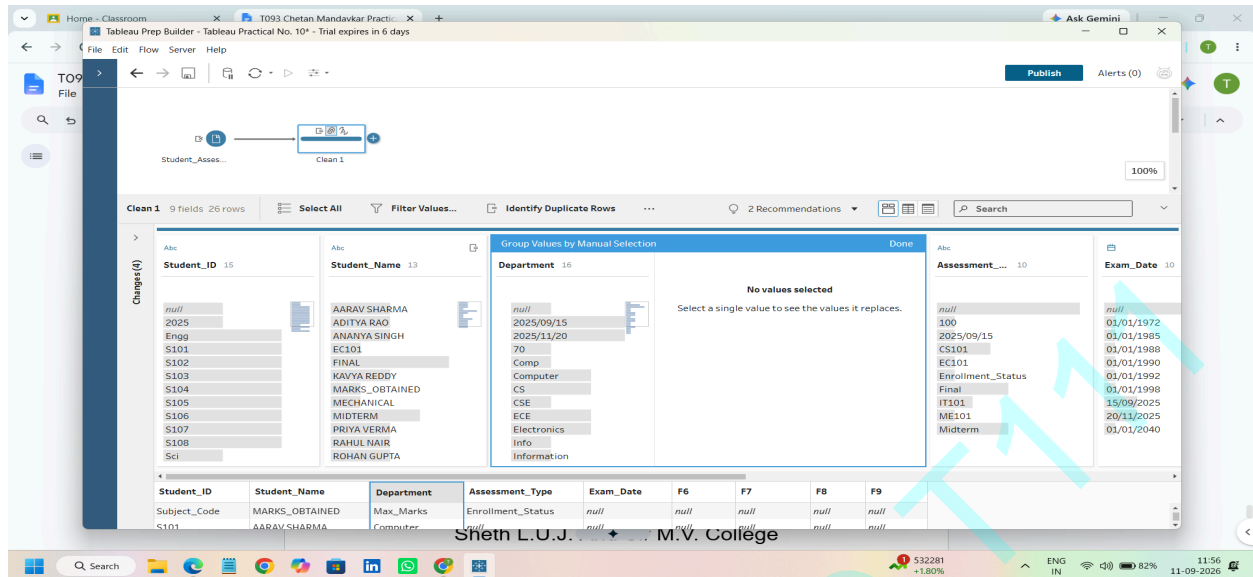


On the Department card, click the More options (...) menu at the top right. Go to Group Values -->select Manual Selection (or Pronunciation / Common Characters).



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Step 2:- In the Grouping editor:

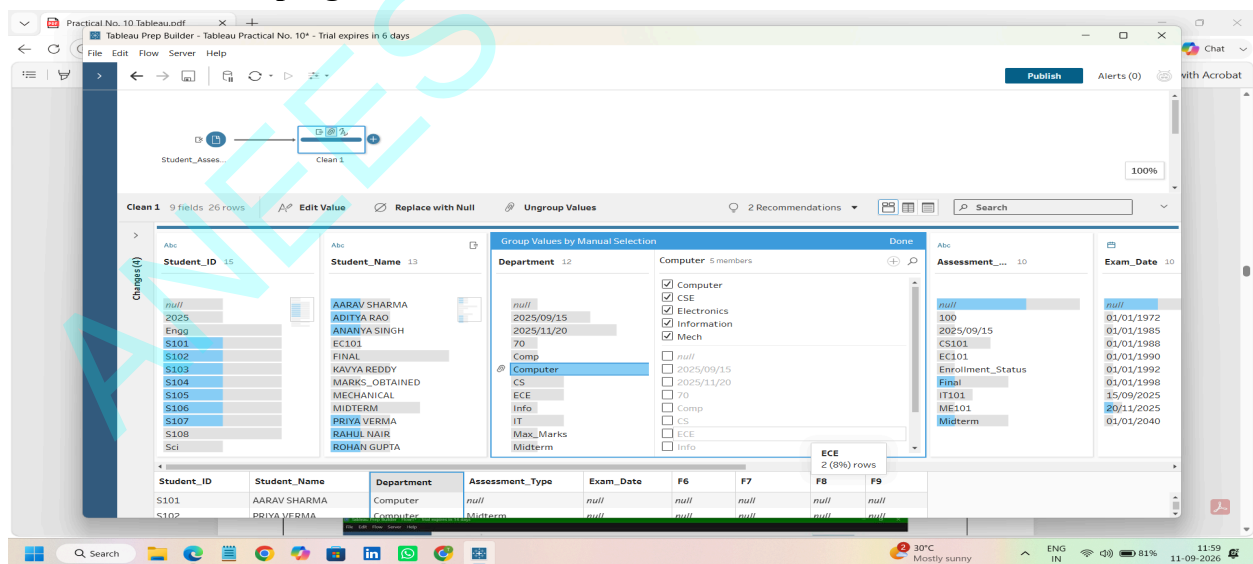
Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE

Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.

Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.

Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.

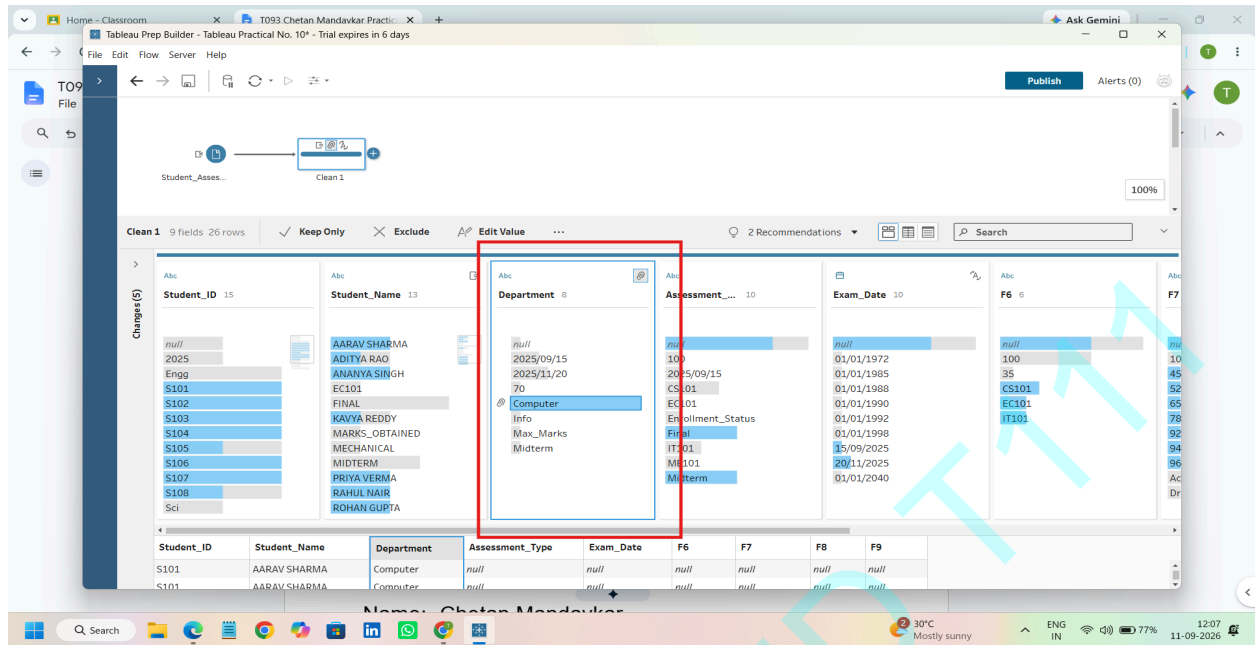
Click Done at the top right of the editor.



Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments.

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Step 3:- Click on the Clean 2 step connected to Student_Assessments_Term2.

2. Locate the Dept card.

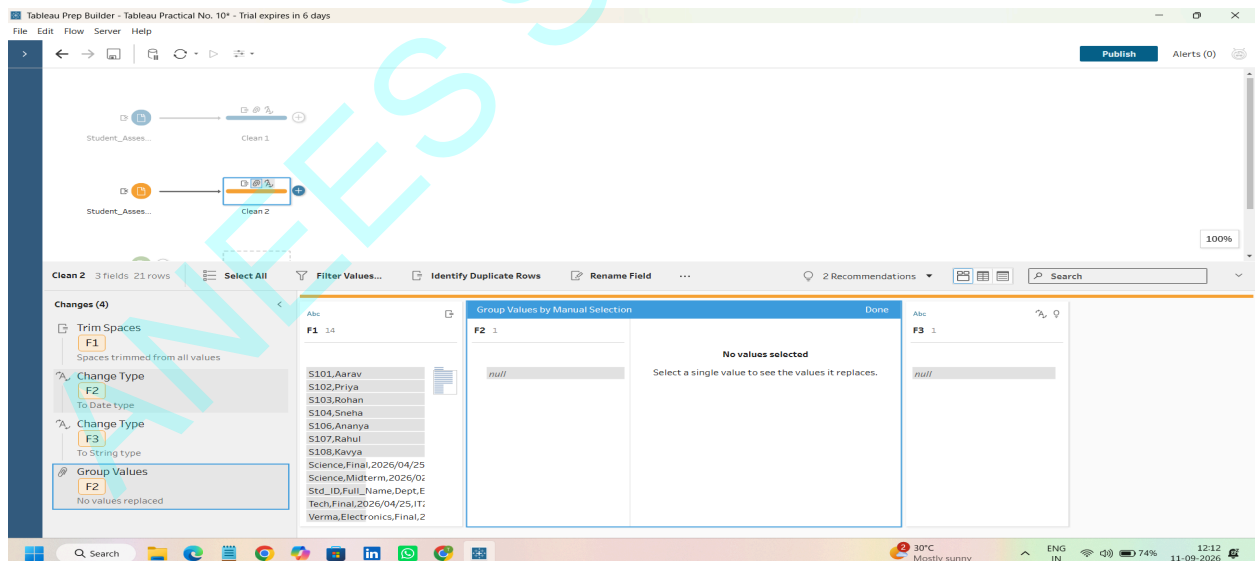
3. Click More options (...) -->Group Values -->Manual Selection:

o Group CS, CSE, and Computer Science into Computer Science.

o Group ECE and Electronics into Electronics.

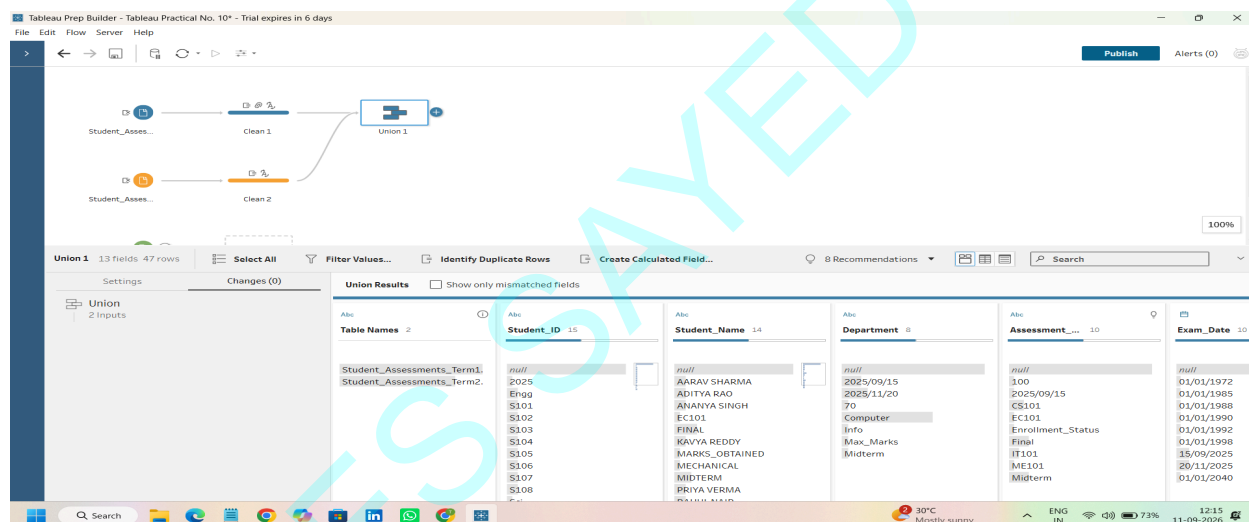
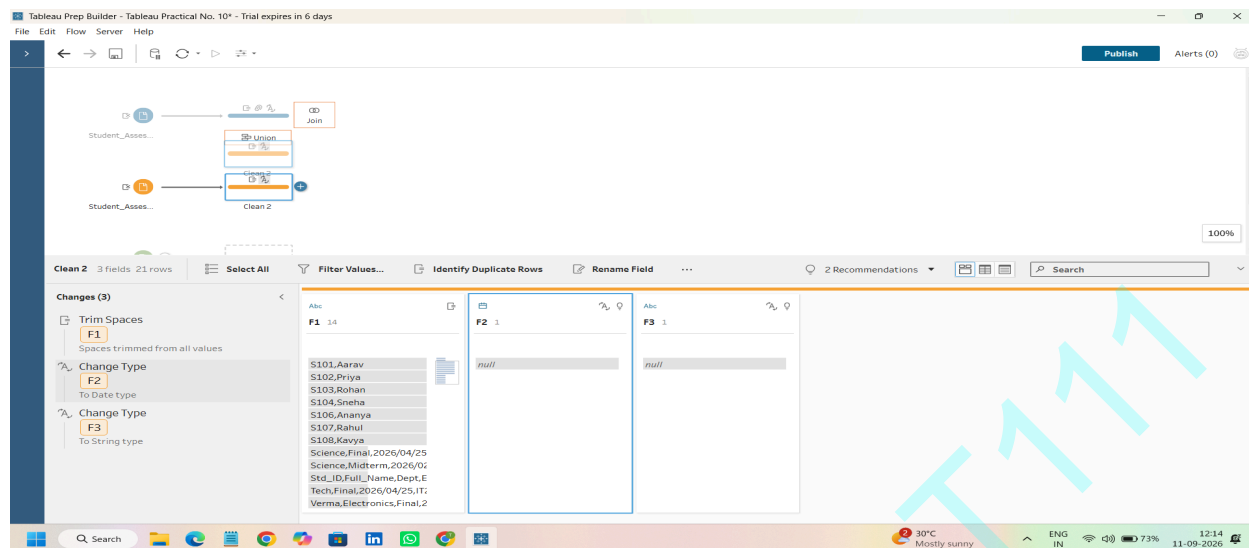
o Group IT and Information Tech into Information Tech.

4. Click Done.



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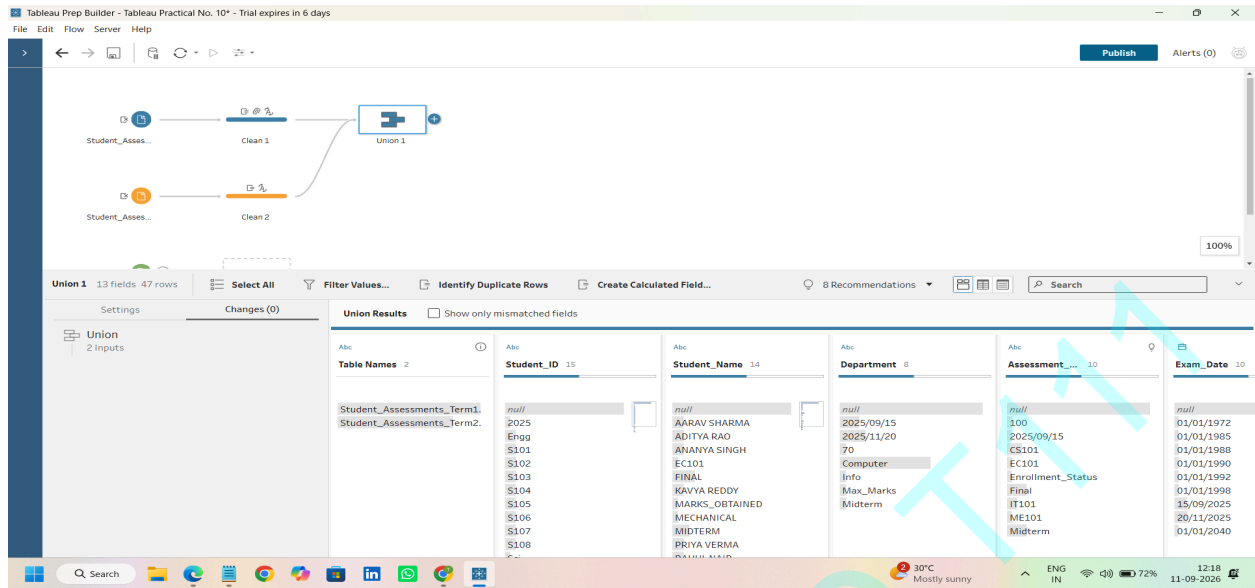
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Step 2:- Click on the Union 1 step. In the bottom Union summary pane, look at the Mismatched Fields section.

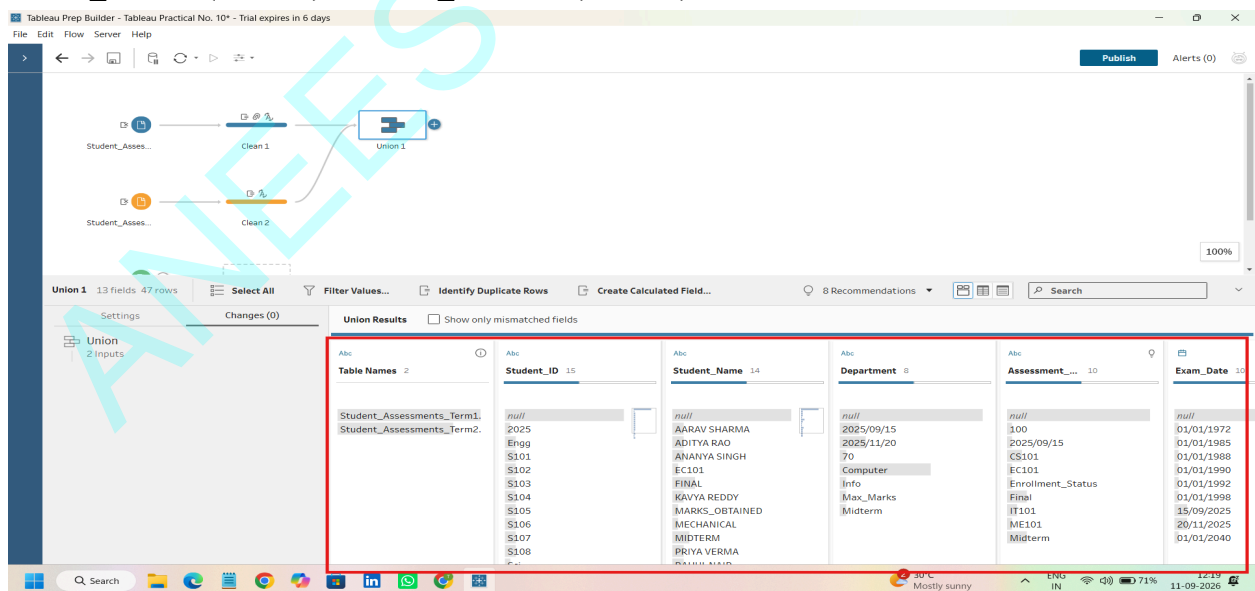
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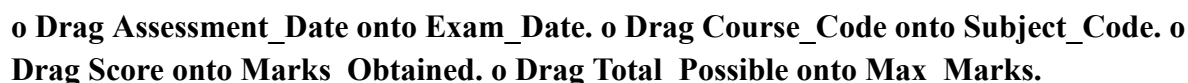
Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:

- o Student_ID (Term 1) vs Std_ID (Term 2)
- o Student_Name (Term 1) vs Full_Name (Term 2)
- o Department (Term 1) vs Dept (Term 2)
- o Assessment_Type (Term 1) vs Exam_Type (Term 2)
- o Exam_Date (Term 1) vs Assessment_Date (Term 2)
- o Subject_Code (Term 1) vs Course_Code (Term 2)
- o Marks_Obtained (Term 1) vs Score (Term 2)
- o Max_Marks (Term 1) vs Total_Possible (Term 2)



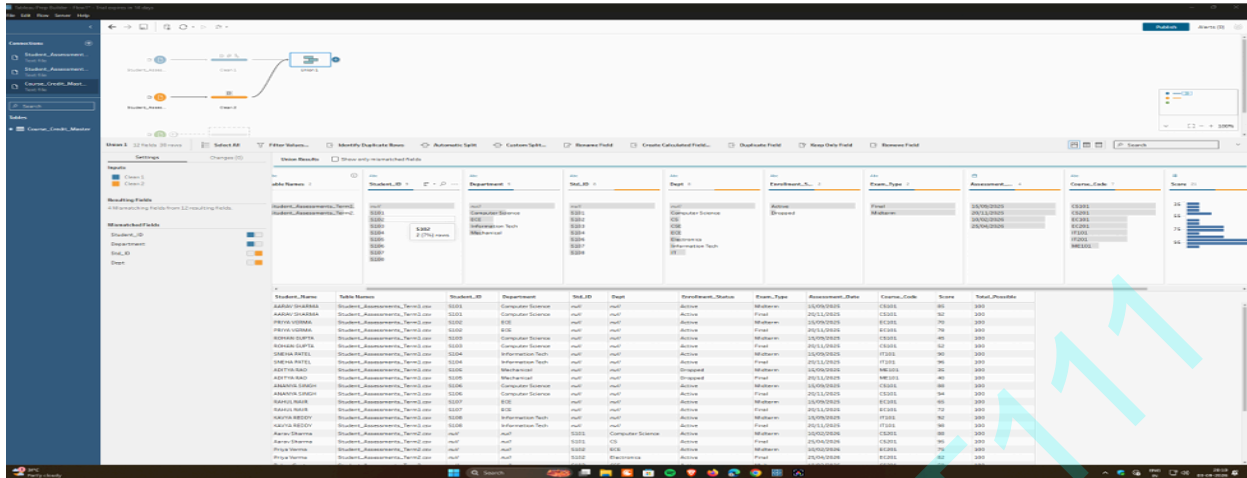
Step 3:- In the Mismatched Fields list (or the union profile pane):

- o Click and drag Std_ID directly on top of Student_ID to merge them into one column. o Drag Full_Name onto Student_Name.
- o Drag Dept onto Department.

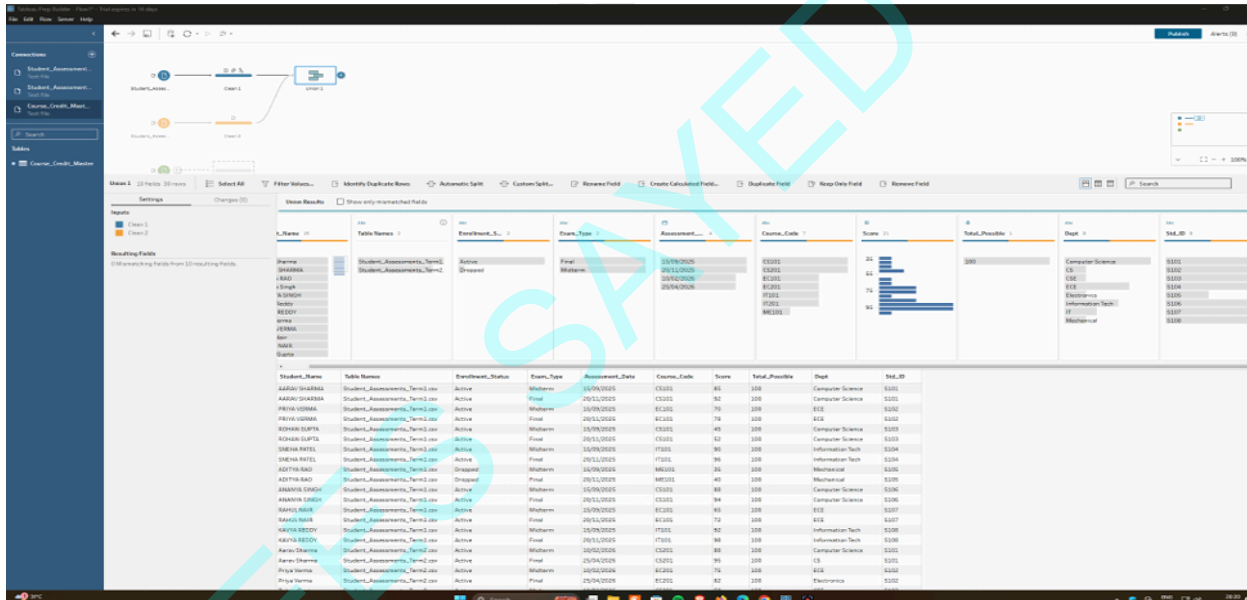


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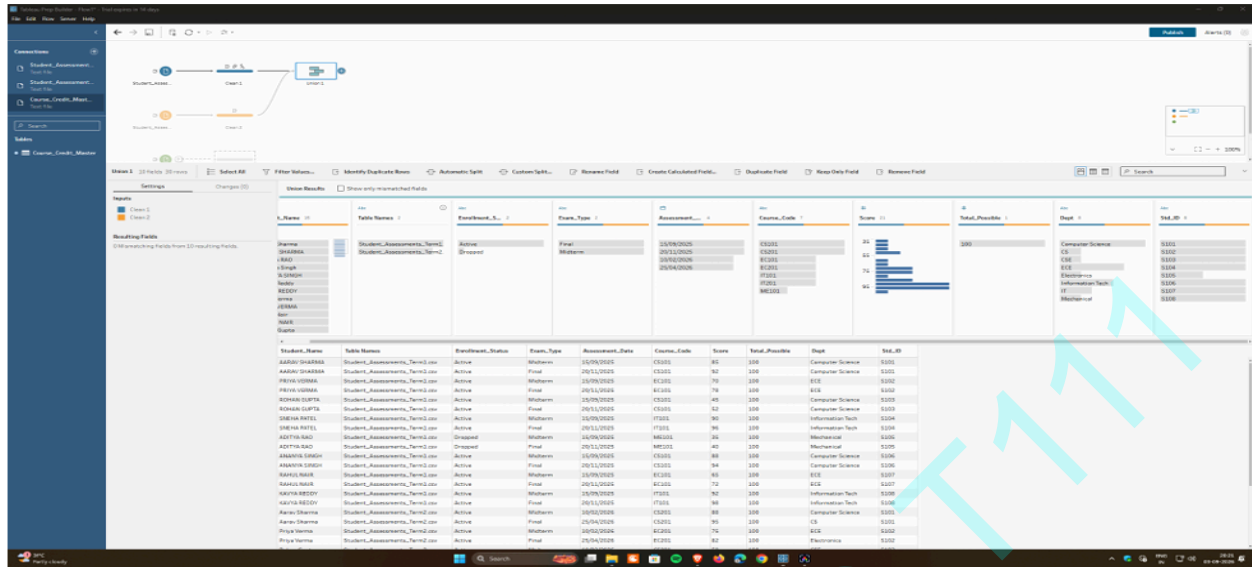
Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.



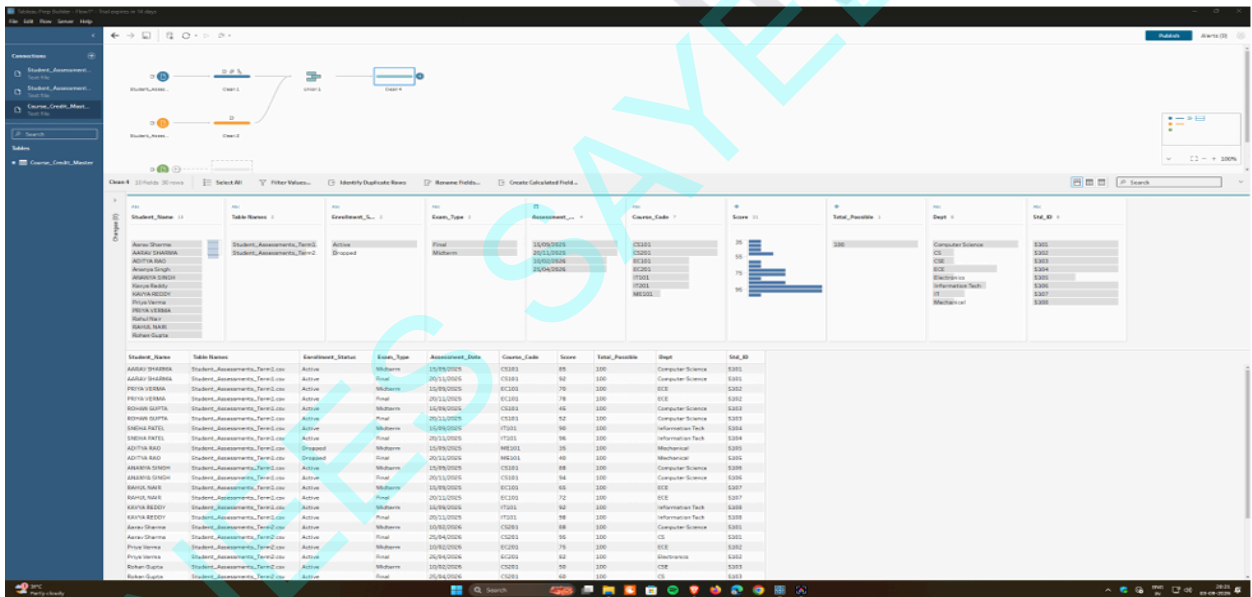
Step 4:- Click the + icon next to the Union 1 node.

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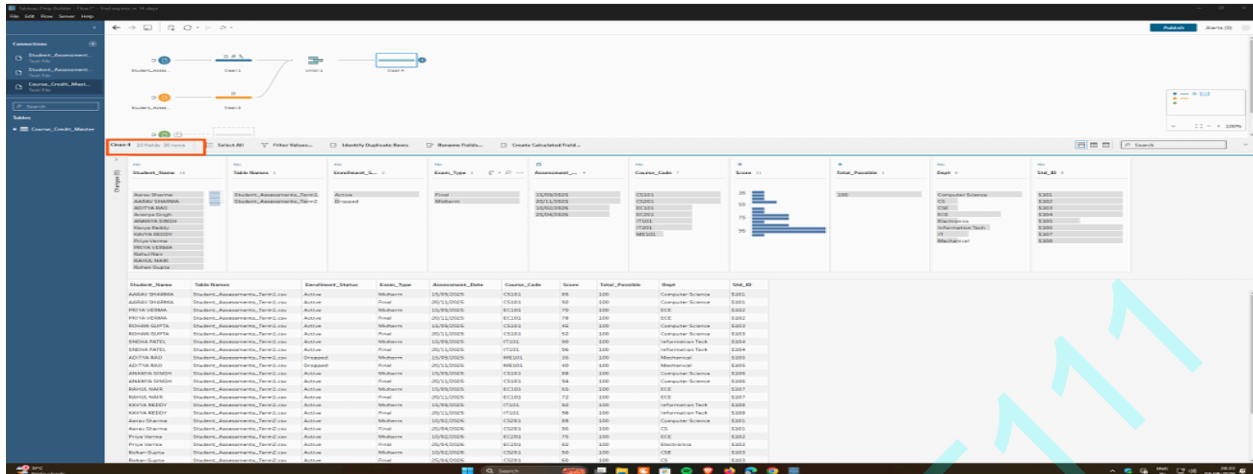
Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.



Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

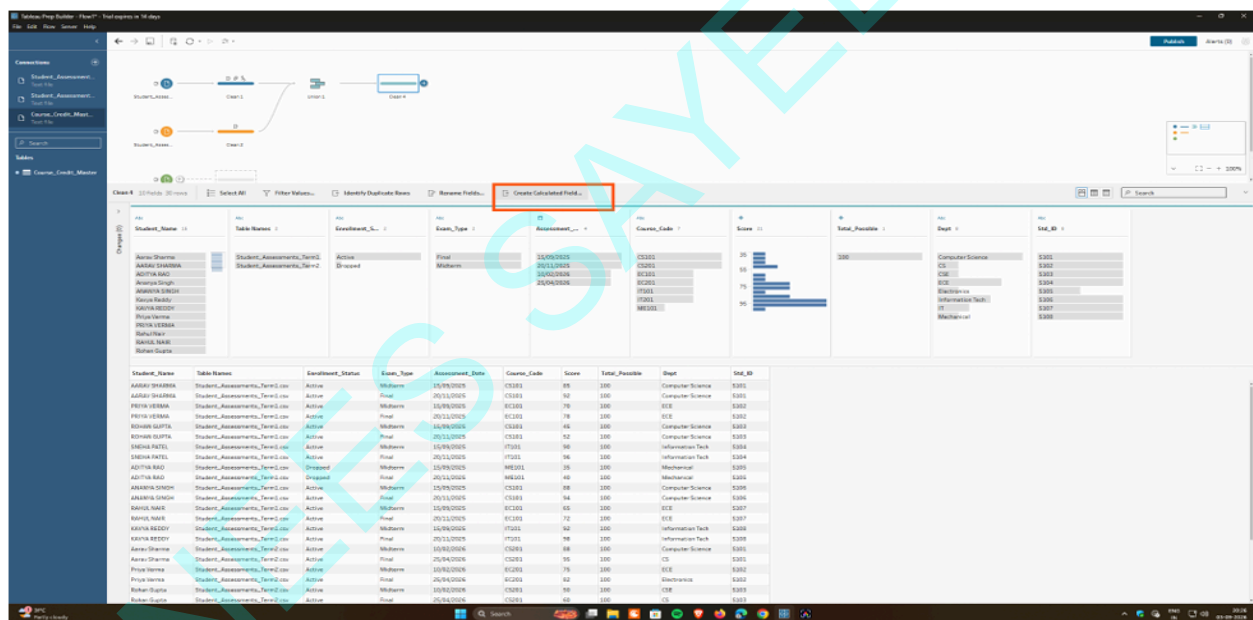
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E: Perform Aggregations in Prep

Step 1:- Click on the Clean 3 step (the clean step created after Union 1). In the toolbar above the Profile cards, click Create Calculated Field You have to click three dots on the toolbar.



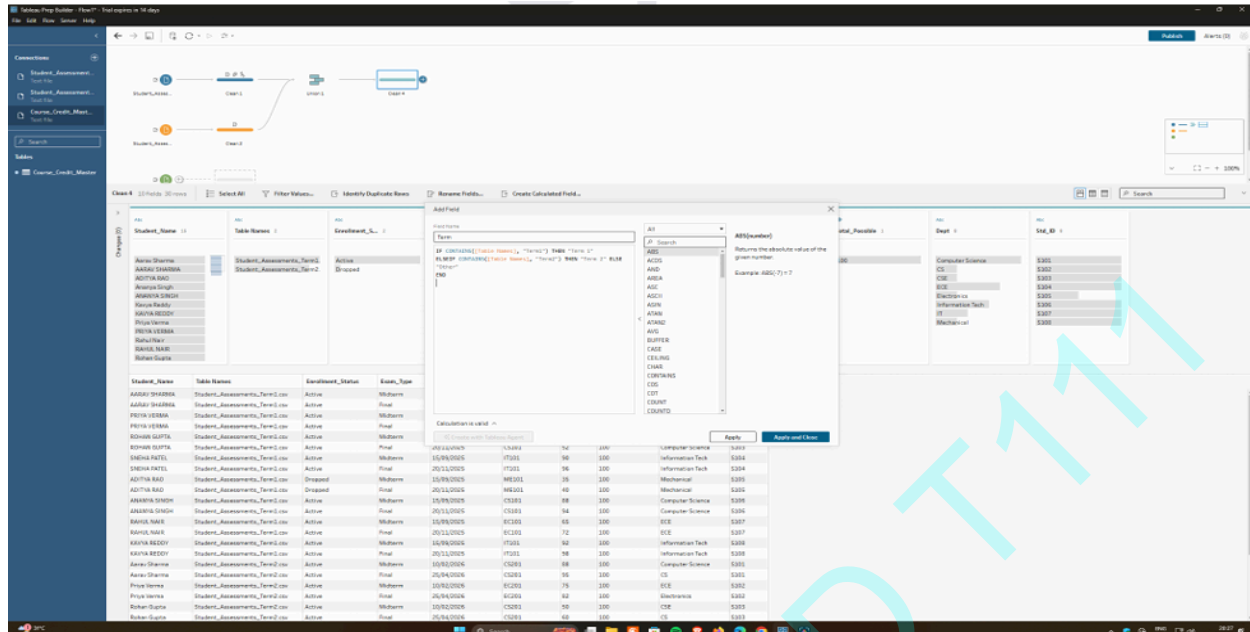
3. Set the Field Name to: Term

4. Enter the formula: Code snippet IF CONTAINS([Table Names], "Term1") THEN "Term 1" ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2" ELSE "Other" END

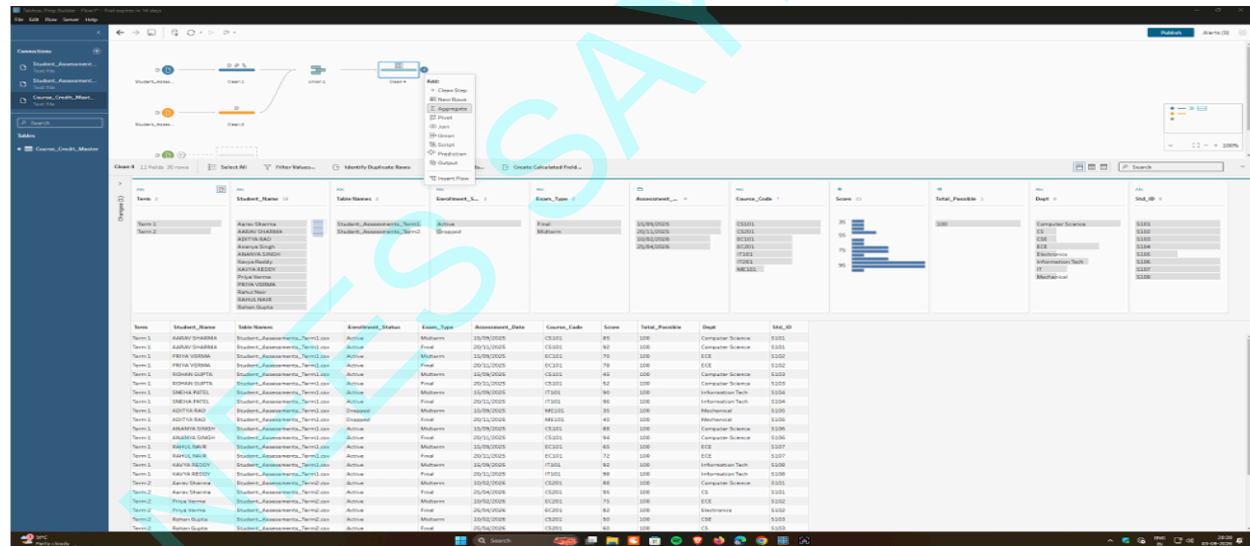
5. Click Save.

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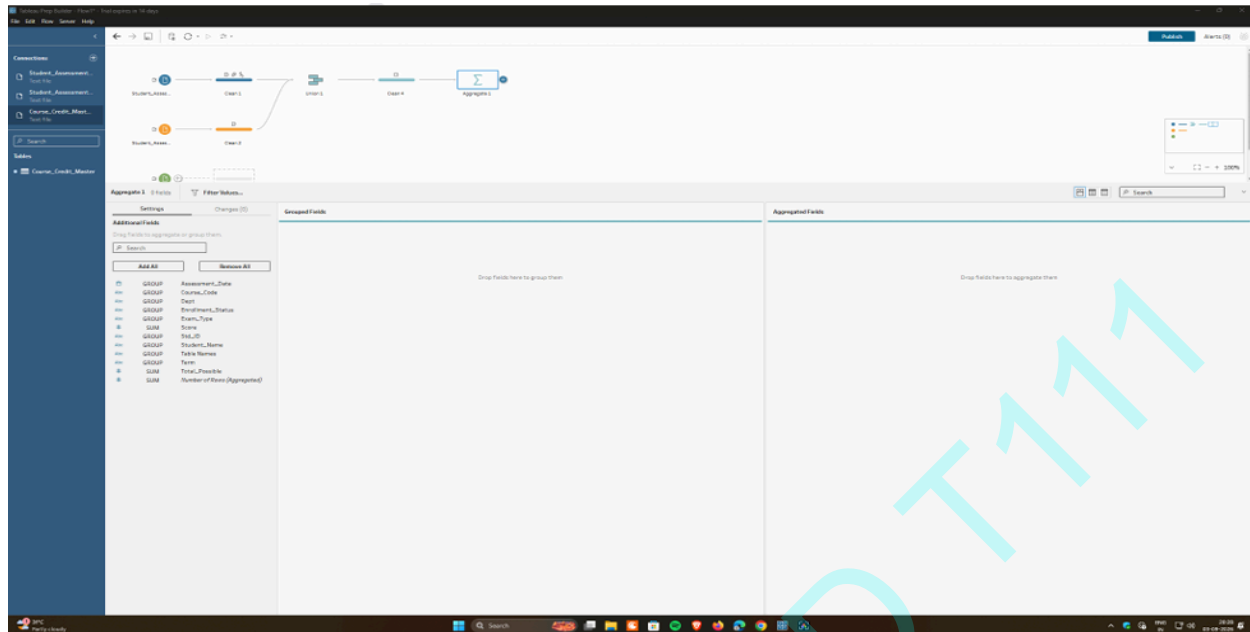
Step 2:- Add an Aggregate Step to the Flow Canvas 1. Hover over the Clean 3 node on the Flow canvas. 2. Click the + icon next to it. 3. Select Aggregate (represented by the summation icon \$\sum\$)



4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane: o Grouped Fields (Left Zone) o Aggregated Fields (Right Zone)

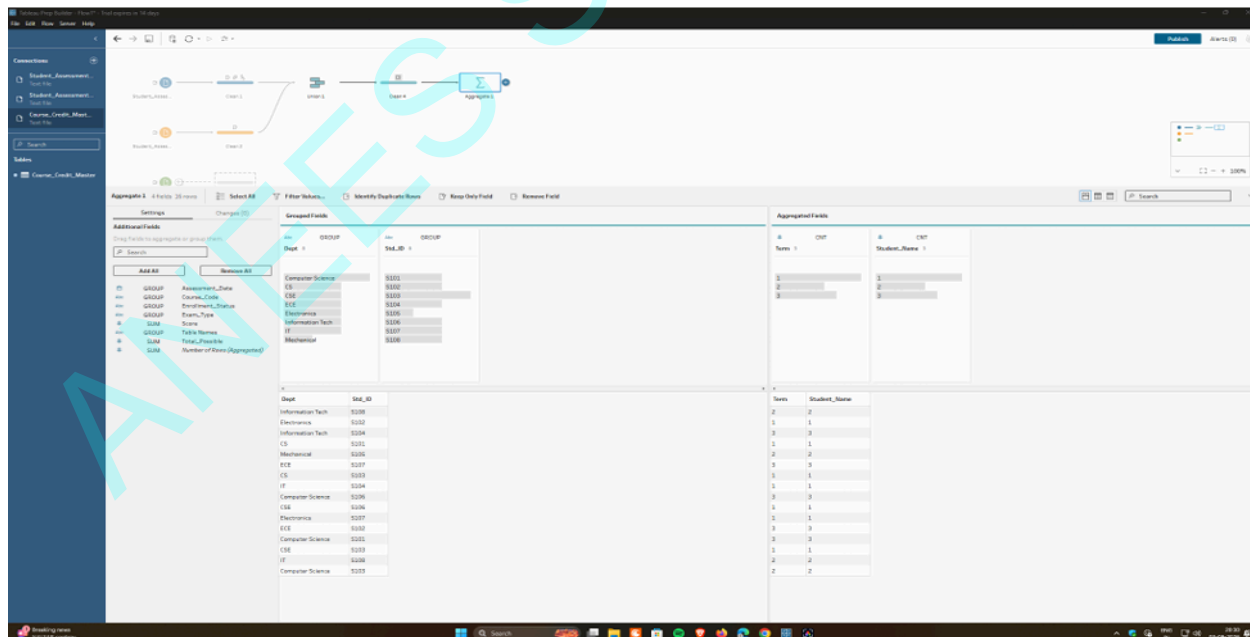
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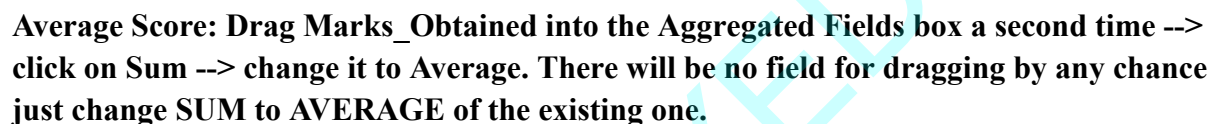
Step 3:- Configure Grouped Fields (Dimensions) From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.



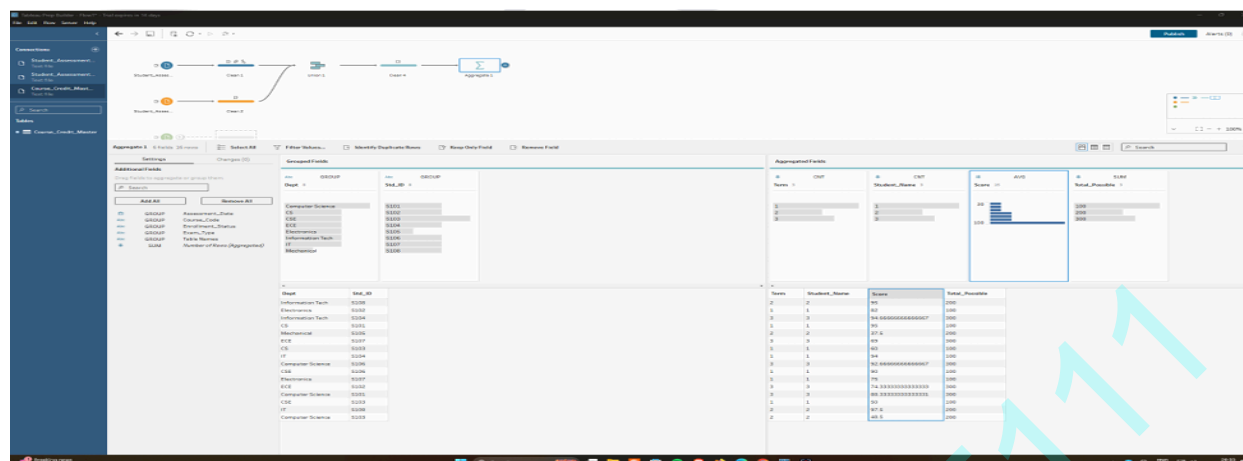
Step 4: Configure Aggregated Fields (Measures & Metrics) 1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure

aggregation operation is set to Sum).

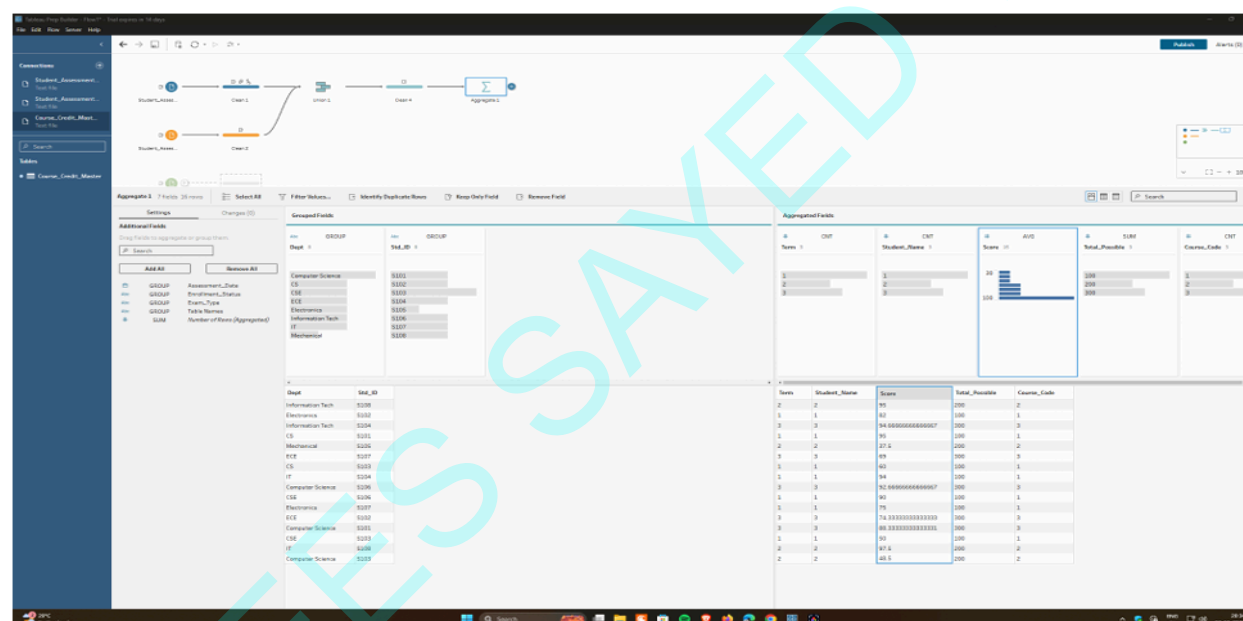


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Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct).



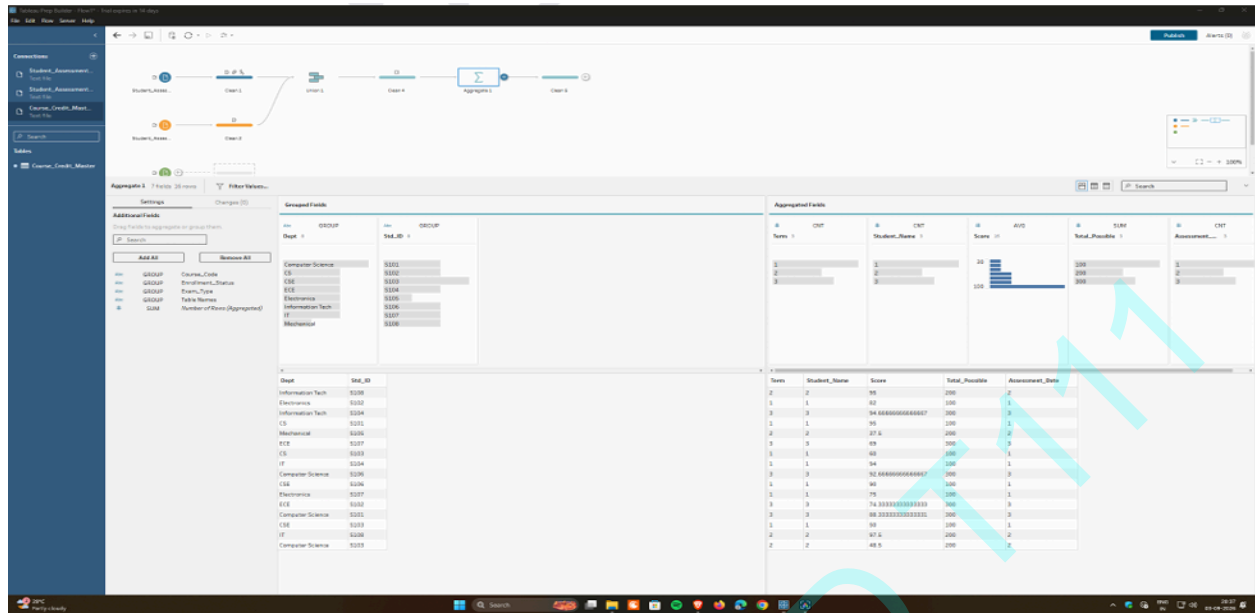
Rename Aggregated Output Fields In the Aggregated Fields panel, click on each field name to rename them cleanly:

- Marks_Obtained (Sum) --> Total_Marks_Obtained
- Marks_Obtained (Average) --> Average_Score
- Max_Marks (Sum) --> Total_Max_Marks
- Assessment_Type (Count) --> Assessments_Taken

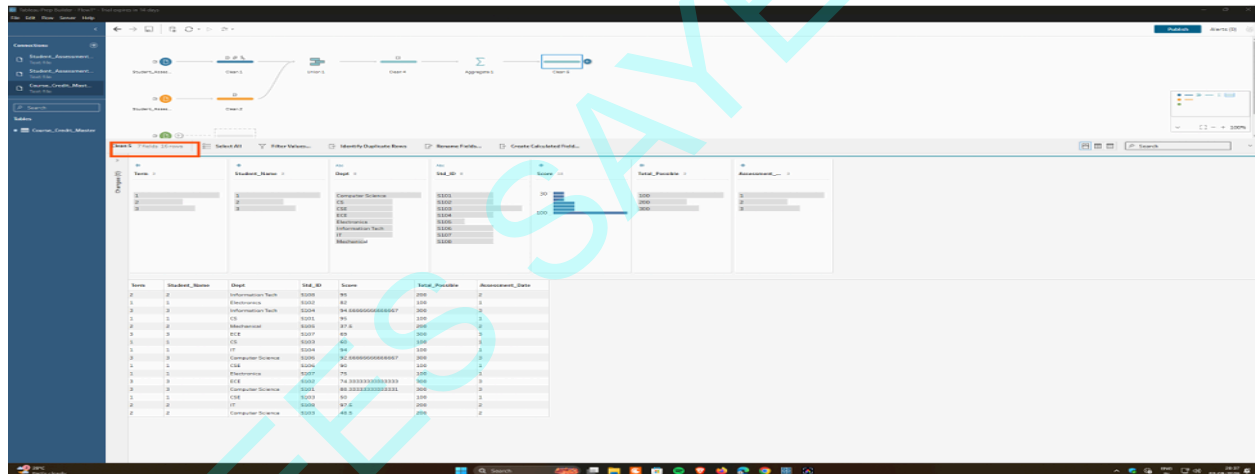
Step 5:- Add a Clean Step to Inspect Aggregated Output 1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean 4).

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Look at the Profile Pane: o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

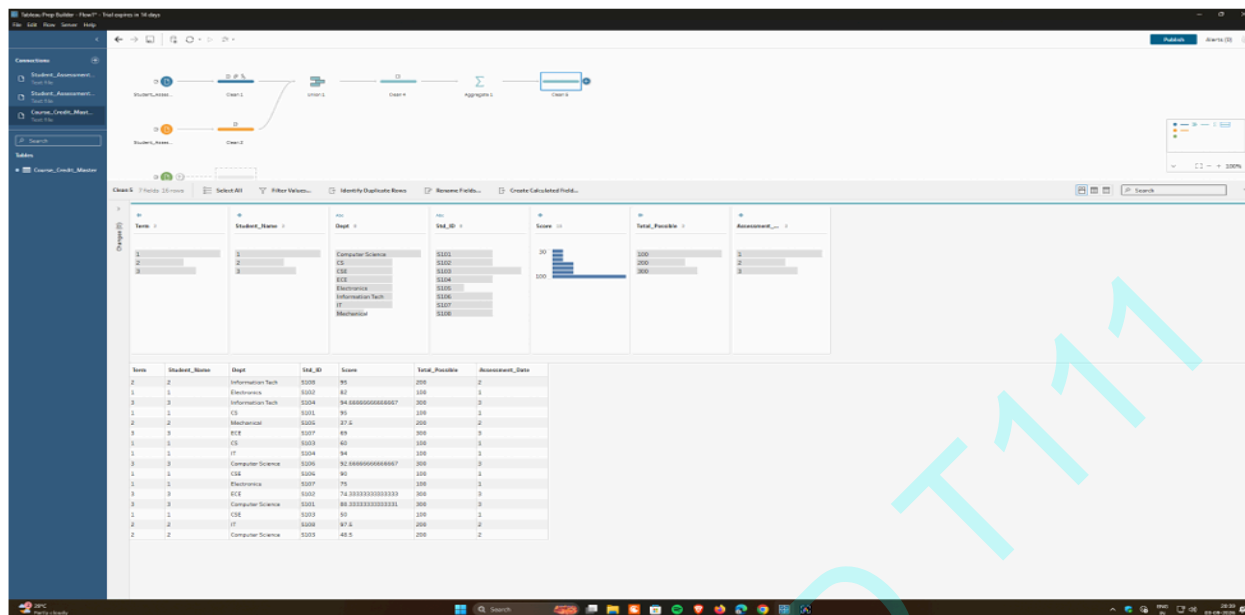


F: Apply Filters and Transformations

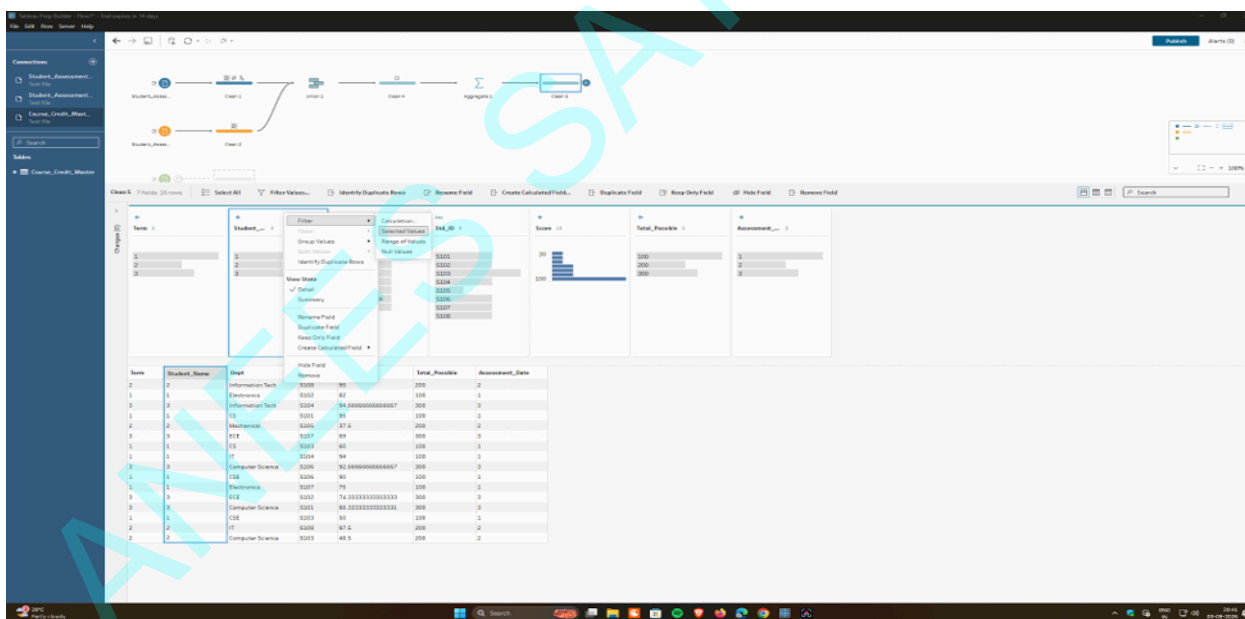
Step 1:- Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

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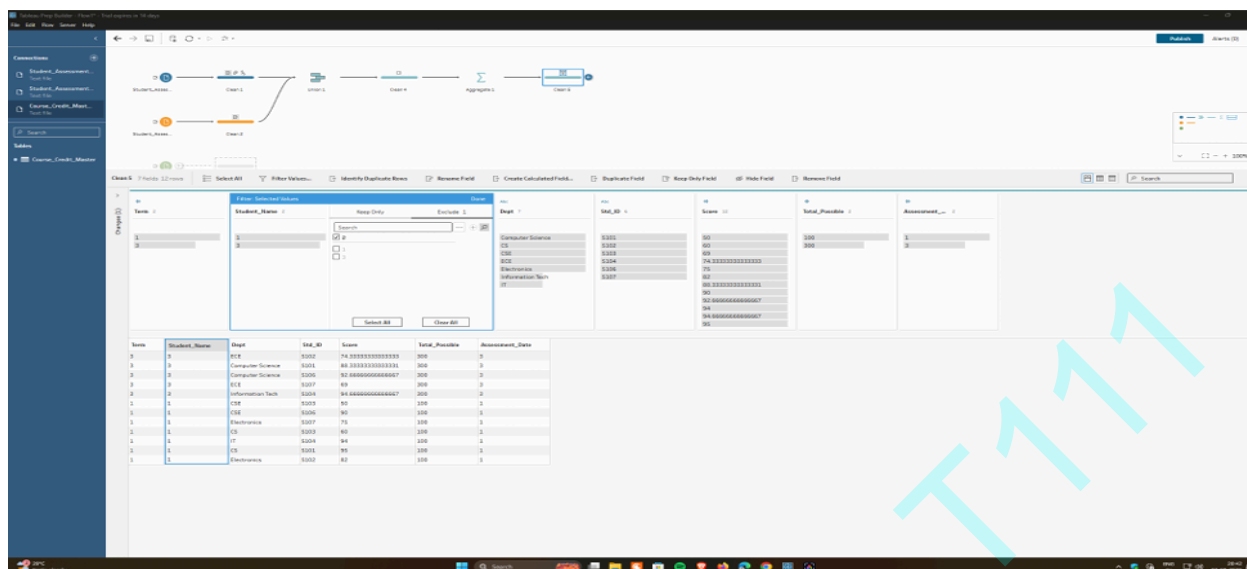
Step 2:- If Enrollment_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active). 2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



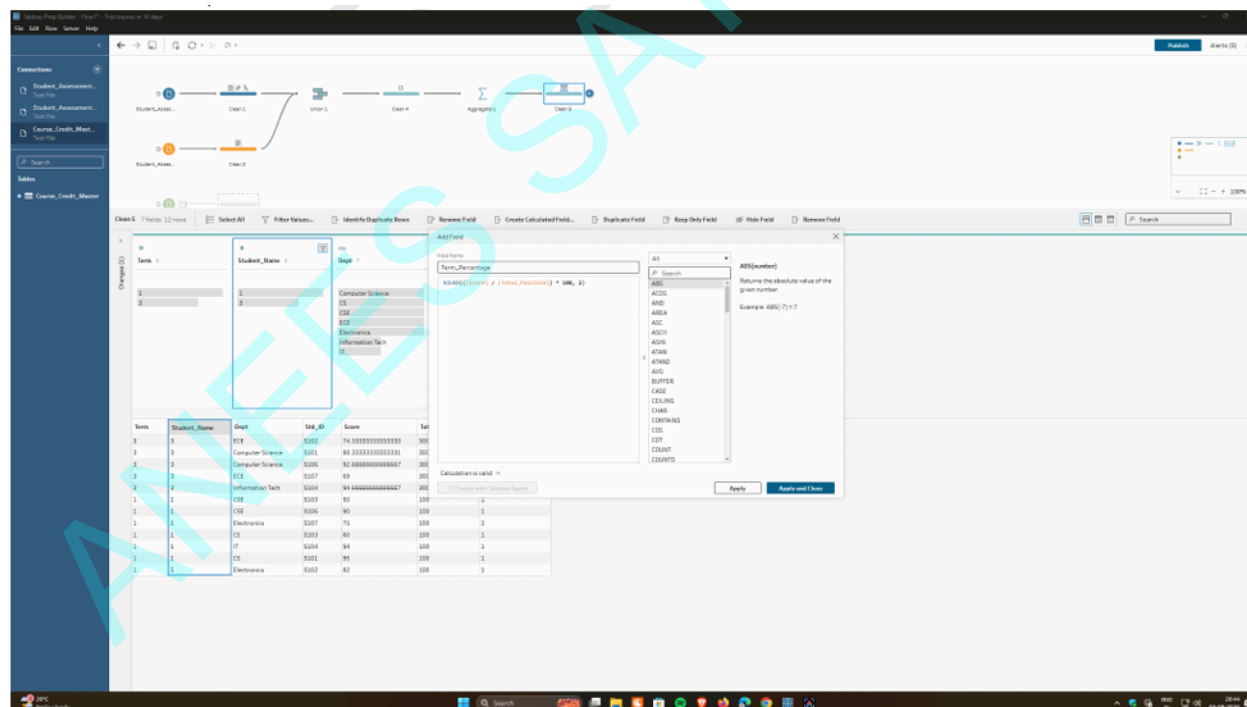
Exclude Dropped so only Active students remain in the active pipeline.

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Step 3:- Create the Normalized Term Percentage Metric 1. In the toolbar above the Profile cards, click Create Calculated Field. 2. Field Name: Term_Percentage 3. Tableau Prep Formula: Code snippet $\text{ROUND}([Total_Marks_Obtained] / [Total_Max_Marks]) * 100, 2$ $\text{ROUND}([Score] / [Total_Possible]) * 100, 2$ {This formula can be used if you couldn't able to rename your aggregate fields.} 4. Click Save.



Step 4;- In the toolbar, click Create Calculated Field.

2. Field Name: Letter_Grade

3. Tableau Prep Formula: Code snippet IF [Term_Percentage] >= 85 THEN "A (Distinction)" ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)" ELSEIF

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[Term_Percentage] >= 60 THEN "C (Second Class)" ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)" ELSE "F (Fail / Re-appear)" END

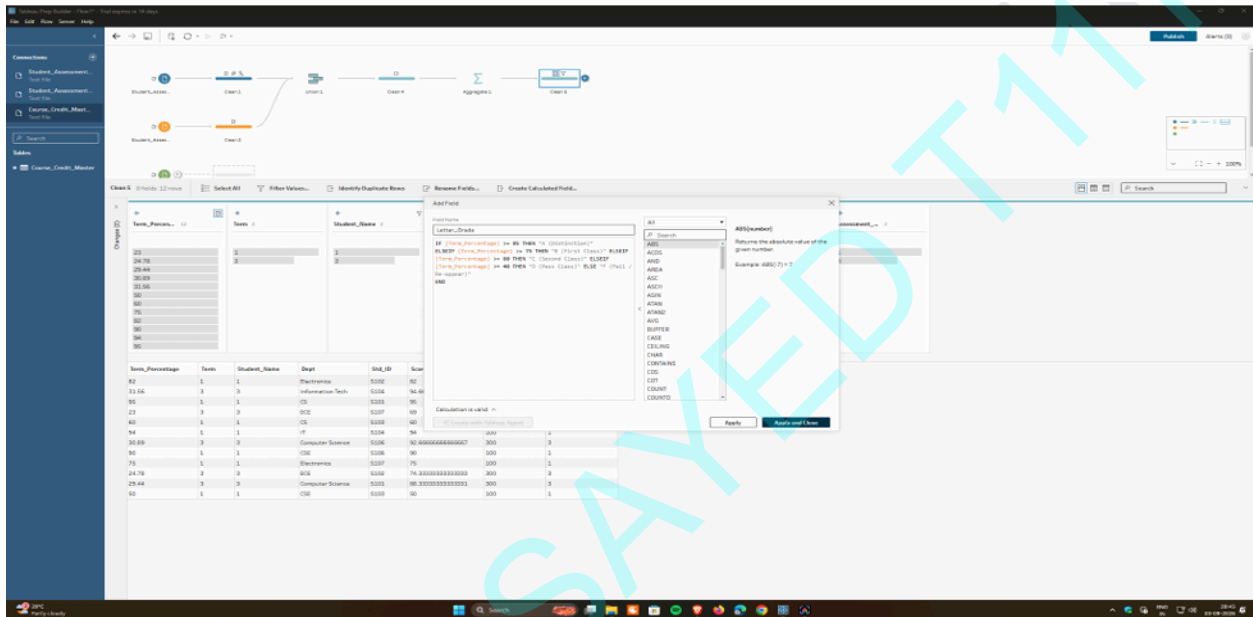
4. Click Save. Step

5: Create Academic Standing Compliance Flag

1. Click Create Calculated Field.

2. Field Name: Academic_Standing

3. Tableau Prep Formula: Code snippet IF [Term_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation (<60%)" END



Step 5:- 1. Click Create Calculated Field.

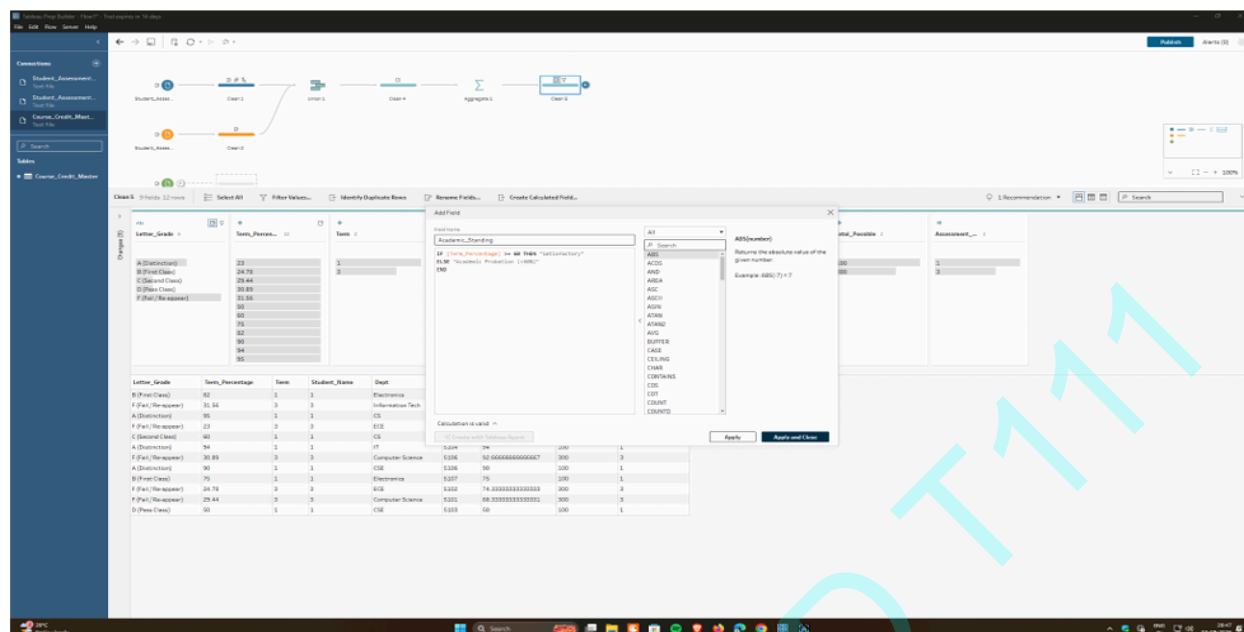
2. Field Name: Academic_Standing

3. Tableau Prep Formula: Code snippet IF [Term_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation (<60%)" END

4. Click Save.

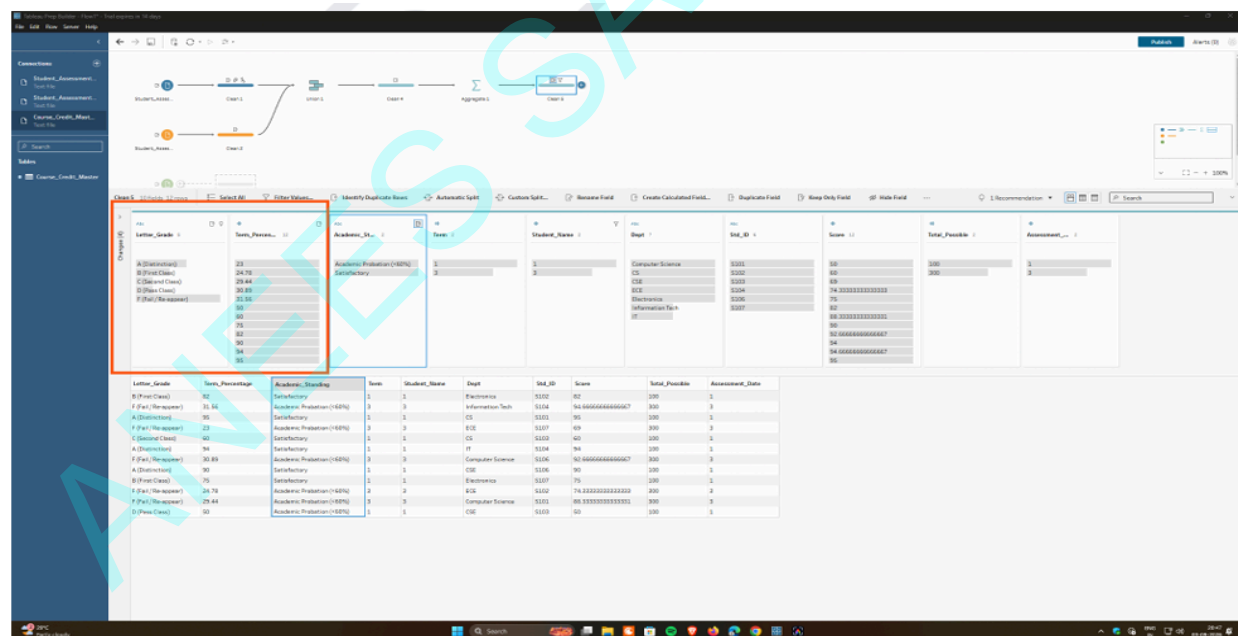
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Step 6:- 1. Look at the newly generated profile cards in Clean 4:

- o **Term_Percentage:** Displays a smooth numeric distribution between 0% and 100%.
- o **Letter_Grade:** Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
- o **Academic_Standing:** Highlights students requiring academic counseling.



G: Create and Automate Data Preparation Flows

Step 1:- 1. On the Flow canvas, locate the final cleaning step (Clean 4).

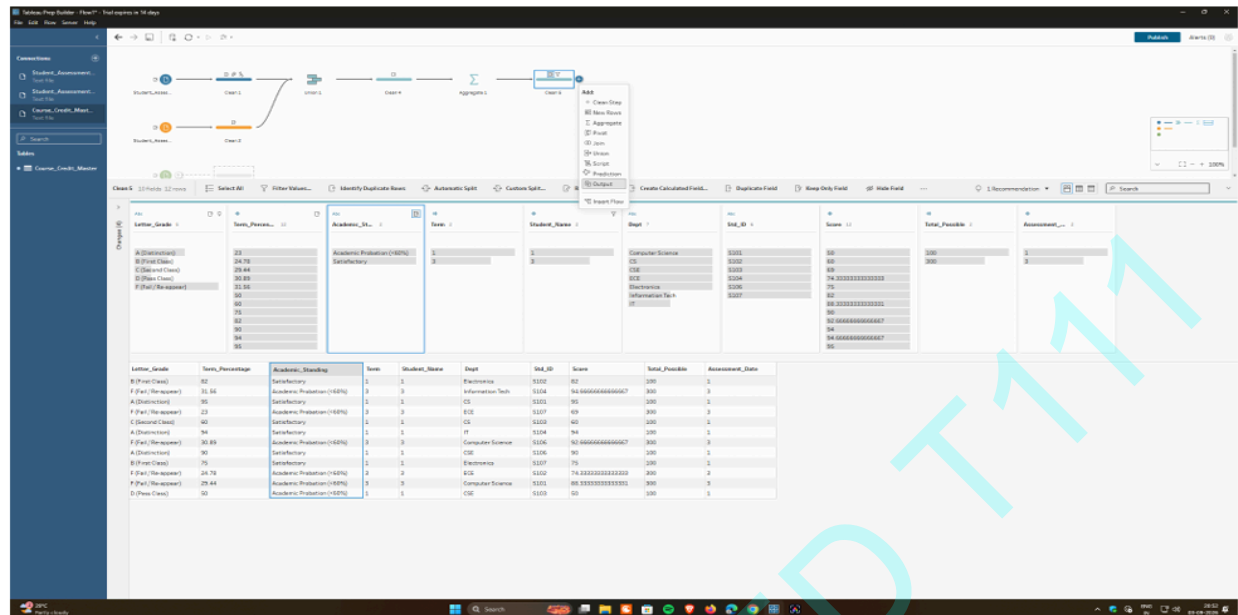
2. Hover over the Clean 4 icon and click the + icon.

3. Select Output (represented by the download/save arrow icon).

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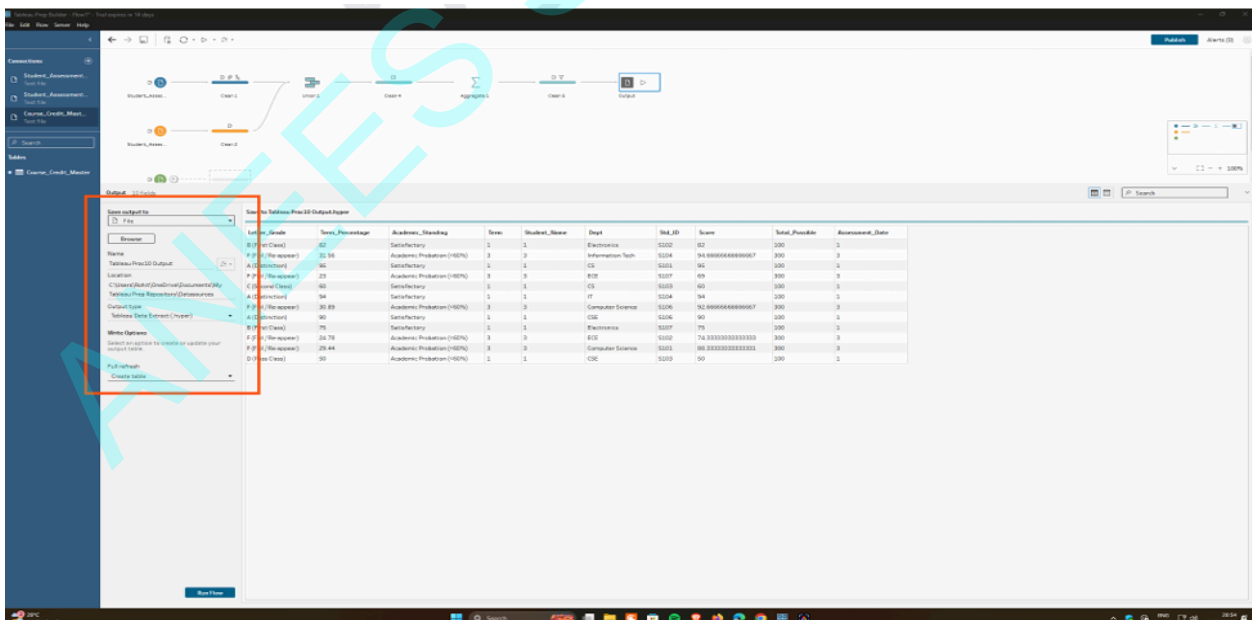
SUBJECT:- TABLEAU

4. An Output step node will appear connected directly to Clean 4



Step 2:- Click on the Output node to open its configuration panel below:

- o Save output to: Select File (or Tableau Server / Tableau Cloud).
- o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
- o Location: Click Browse and choose the directory where you want the final dataset saved.
- o File Name: Enter Student_Performance_Master_Cleaned.
- o Write Option: Select Create table (or Append to table if updating existing records).



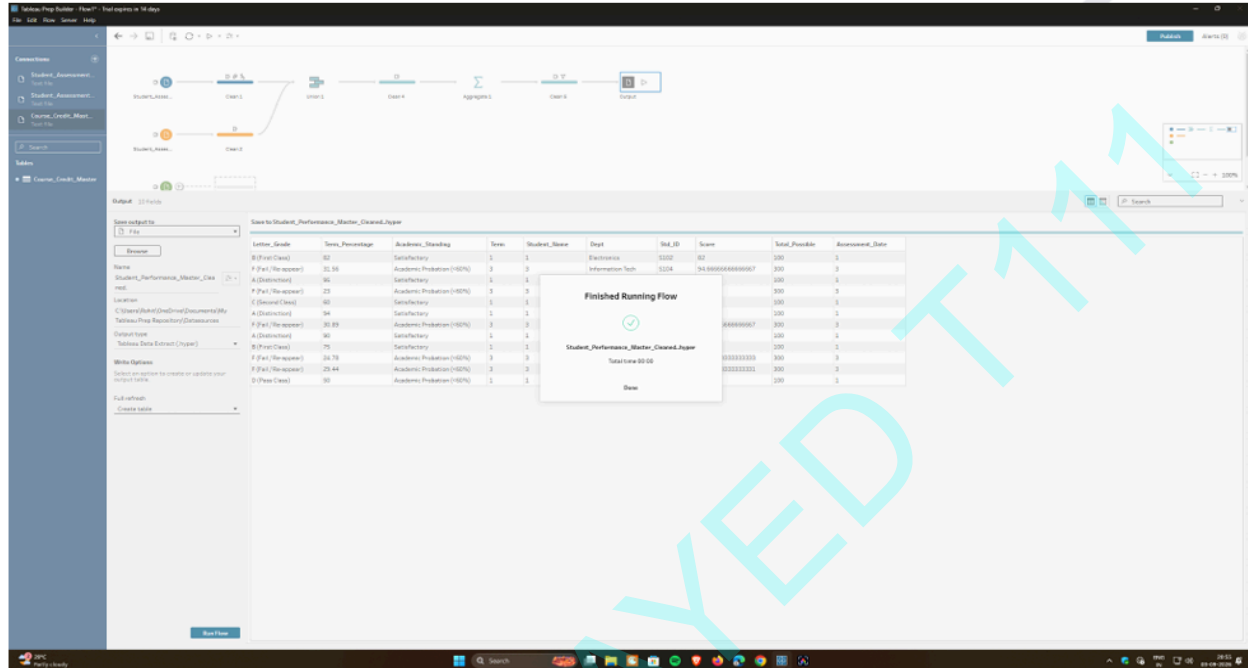
Step 3:- 1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon directly on the Output node on the canvas).

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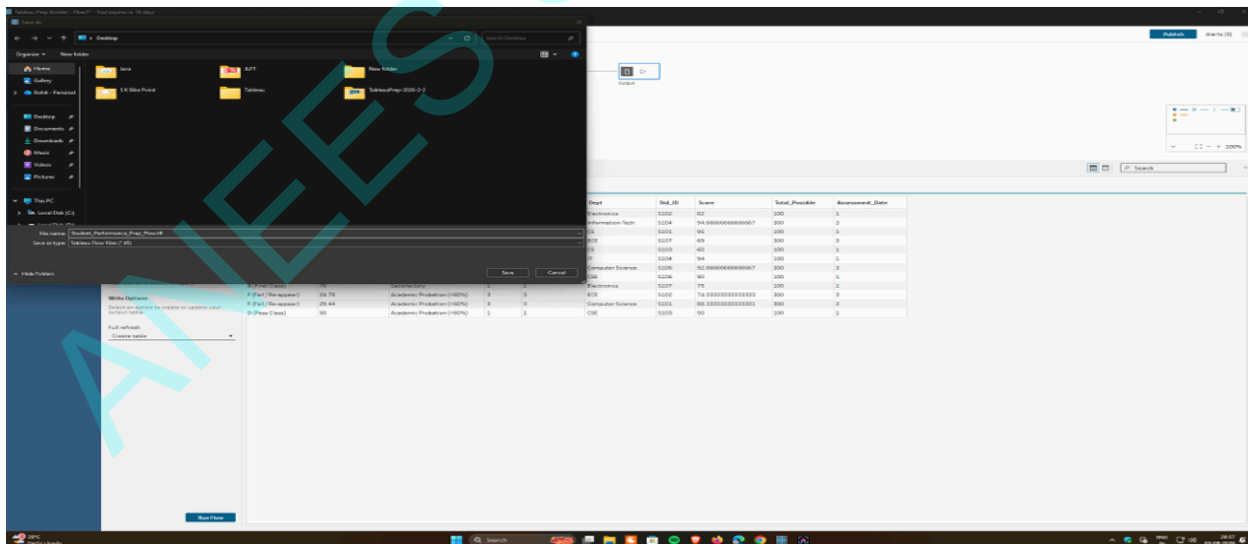
SUBJECT:- TABLEAU

2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).

3. A success notification will appear stating: "The flow ran successfully in X seconds."



Step 4: Save and Automate the Flow File 1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As.... 2. Name your workflow: Plaintext Student_Performance_Prep_Flow.tfl

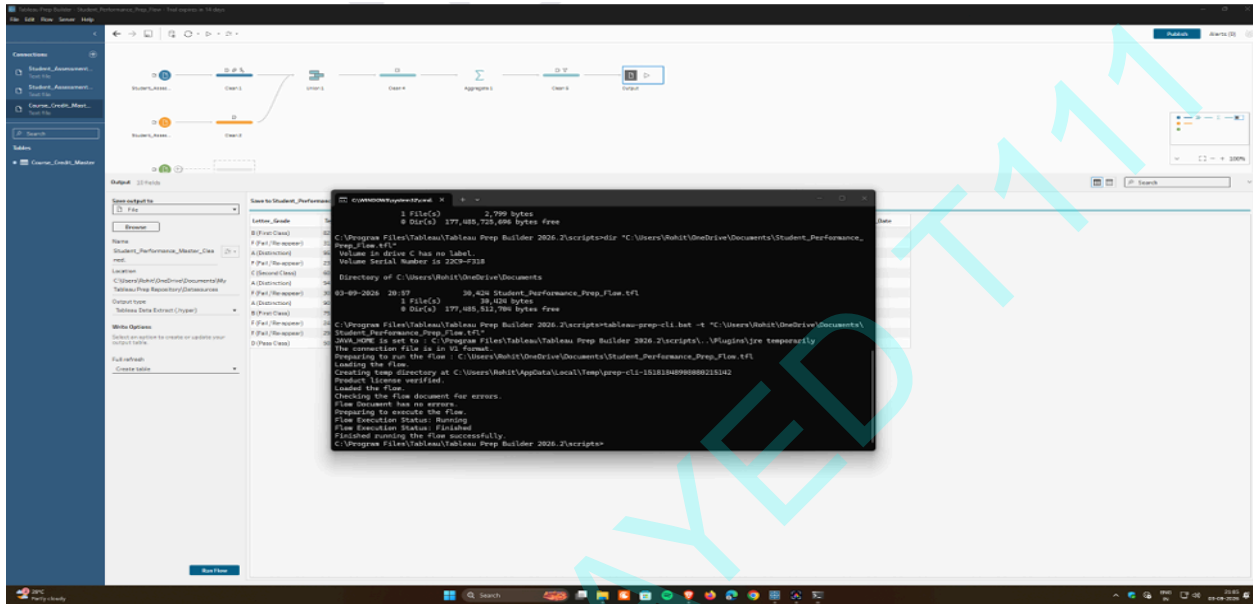


Step 5:- (Automation Options in Production):

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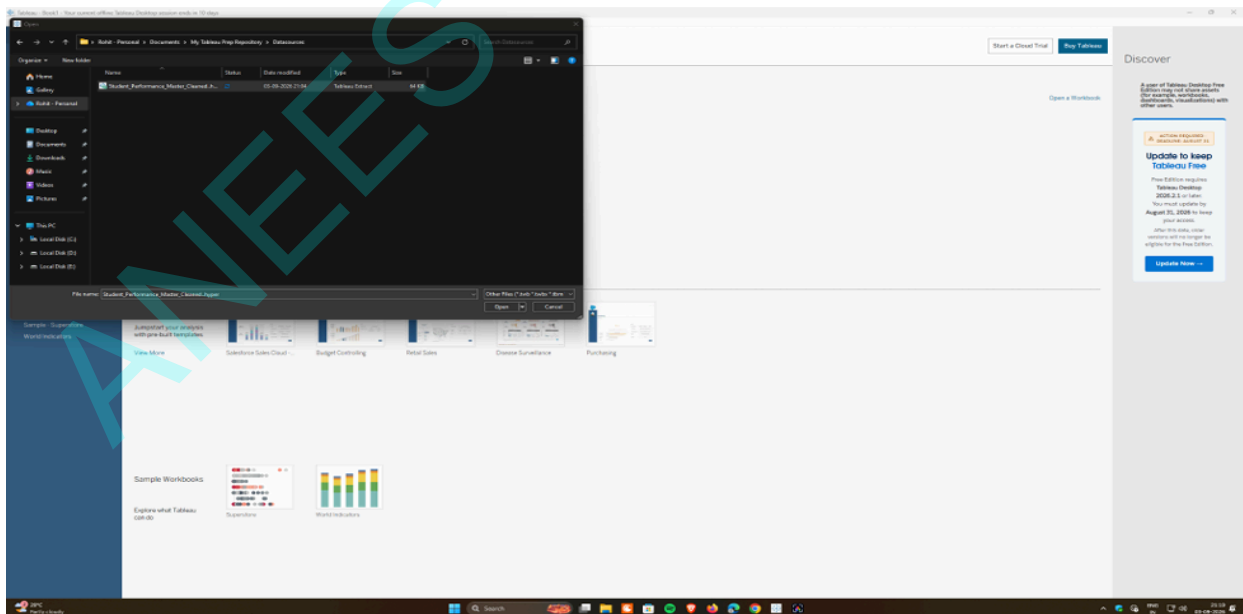
SUBJECT:- TABLEAU

- o **Command Line Execution:** You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI): `Bash tableau-prep-cli -t Student_Performance_Prep_Flow.tfl`
- o **Tableau Prep Conductor:** If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.



Step 6:- Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported `Student_Performance_Master_Cleaned.csv` (or .hyper).



Build the self-performance dashboard using the pre-calculated metrics:

- o Drag `Student_Name` to Rows.

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- o Drag Term to Columns.
- o Drag Term_Percentage to Text.
- o Drag Letter_Grade to Color.
- o Drag Academic_Standing to Tooltip.

